UNIT - 1

Single Degree Freedom Systems

Subject: Mechanical Vibrations | Code: 23A03702a

Syllabus: Un-damped free vibration: classical method, energy method, equivalent systems, torsional systems. Damped free vibration: viscous (under, critical, over damping), Coulomb damping, equivalent damping coefficient. Whirling of shafts: Dunkerley's lower bound, critical speed.

1. Basic Concepts of Vibration

Vibration = repetitive to-and-fro motion of a body (or system) about its mean / equilibrium position.

- *Period (T) : time for one complete cycle, seconds. Frequency (f) = 1/T (Hz).*
- *Circular frequency (ω) = 2πf = 2π/T rad/s. Amplitude (X) = max. displacement from mean position.*
- *Natural frequency (ω_n) : frequency at which a system vibrates by itself after an initial disturbance, with no external force.*
- *Degrees of freedom (DOF) : minimum number of independent co-ordinates required to completely define the position of the system at any instant.*
  - *→ Single DOF → one co-ordinate (x or θ). Two DOF → two co-ordinates, and so on.*
- *Simple Harmonic Motion (SHM) : x = X sin ωt ; velocity = ωX cos ωt ; acceleration = $-\omega^2 x$ (acceleration ~ displacement, directed to mean position).*

Classification of vibration

Basis	Types
External force	**Free** (no force after the initial disturbance) / **Forced** (periodic force acts continuously)
Damping	**Undamped** (no energy loss) / **Damped** (energy dissipated by friction etc.)
Governing equation	**Linear** (superposition holds) / **Non-linear**
Nature of excitation	**Deterministic** (predictable) / **Random** (only statistical description)

Motion	**Longitudinal, Transverse, Torsional**

Elements of a vibrating system



Spring: F = k x
k = stiffness (N/m)

Damper: F = c ẋ
c = damping coeff. (N·s/m)

Mass: F = m ẍ
inertia element (kg)

2. Un-damped Free Vibration of SDOF System

A mass m attached to a spring of stiffness k, free to move along one direction (x).

Disturb it and release - it vibrates at its natural frequency.

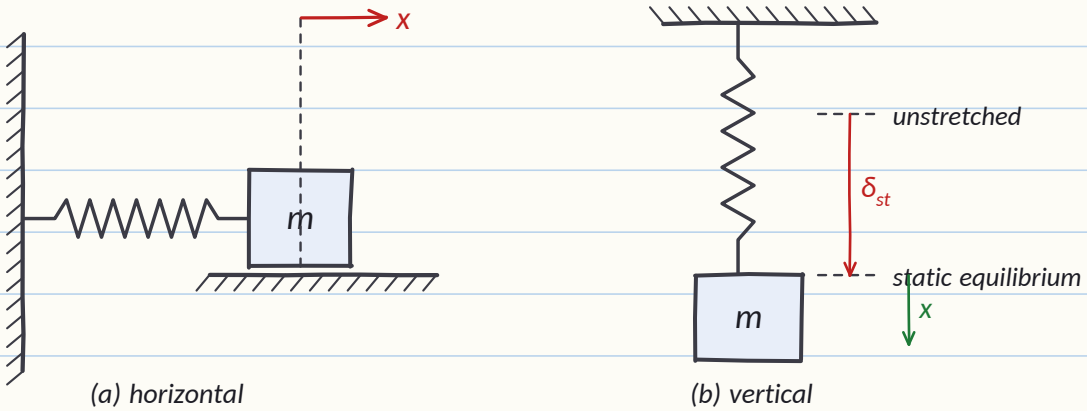


Fig 1.1 Spring-mass system

(a) Classical (Newton's / D'Alembert's) method

Take displacement x from static equilibrium position. In vertical system, spring force at equilibrium balances weight: $k\,\delta_{st} = m\,g$ (so gravity drops out of the equation).

Newton's 2nd law: $m\,\ddot{x} = -k\,x$

$$m\,\ddot{x} + k\,x = 0 \text{ or } \ddot{x} + \omega_n^2\,x = 0 \quad (1)$$

Try $x = e^{st} \rightarrow m\,s^2 + k = 0 \rightarrow s = \pm i\,\sqrt{(k/m)} = \pm i\,\omega_n$

$$\omega_n = \sqrt{(k/m)} \text{ rad/s} = \sqrt{(g/\delta_{st})}$$
$$f_n = (1/2\pi)\sqrt{(k/m)} \text{ Hz } T = 2\pi\sqrt{(m/k)} \text{ s} \quad (2)$$

General solution: $x(t) = A\cos\omega_n t + B\sin\omega_n t$

With initial conditions $x(0) = x_0$, $\dot{x}(0) = \dot{x}_0$:

$$x(t) = x_0\cos\omega_n t + (\dot{x}_0/\omega_n)\sin\omega_n t = X\cos(\omega_n t - \varphi)$$
$$X = \sqrt{(x_0^2 + (\dot{x}_0/\omega_n)^2)}\ \varphi = \tan^{-1}[\,\dot{x}_0 / (x_0\,\omega_n)\,] \quad (3)$$

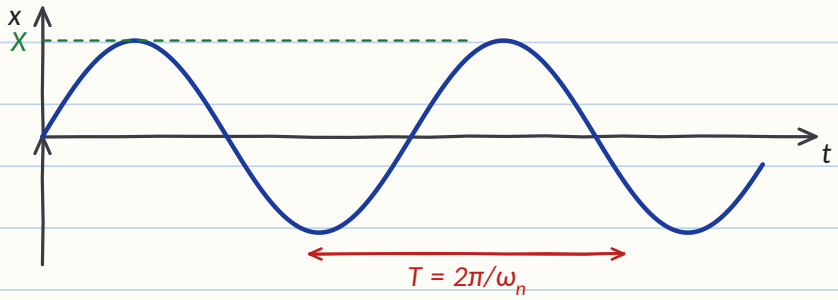


Fig 1.2 Un-damped free response (simple harmonic)

Remember:

Natural frequency depends ONLY on k and m (system properties), not on the initial conditions. Initial conditions decide only amplitude X and phase φ.

(b) Energy method (Rayleigh's method)

For a conservative system total energy is constant: $T + U = \text{constant}$ (T = kinetic, U = potential)

$$d/dt\,(T + U) = 0 \rightarrow \text{equation of motion}$$
$$\text{or } T_{max} = U_{max} \rightarrow \omega_n \quad (4)$$

- For SHM $T_{max} = \tfrac{1}{2}\, m_{eff}\, (\omega_n X)^2$, $U_{max} = \tfrac{1}{2}\, k_{eff}\, X^2 \rightarrow \omega_n = \sqrt{(k_{eff}/m_{eff})}$.
- Steps: (1) choose co-ordinate, (2) write T and U in terms of it, (3) differentiate or equate maxima.

Spring-mass: $T = \tfrac{1}{2}\, m\, \dot{x}^2$, $U = \tfrac{1}{2}\, k\, x^2$. $d/dt(T+U) = (m\,\ddot{x} + k\,x)\,\dot{x} = 0 \rightarrow m\,\ddot{x} + kx = 0$ (same as Newton).

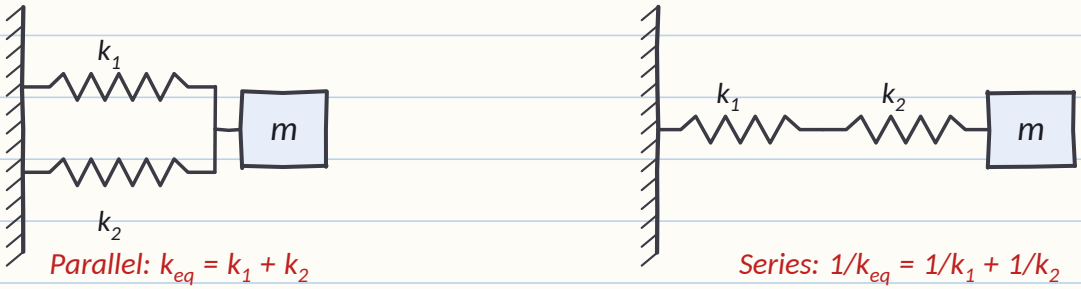
Effect of spring mass (m_s): a small element dy of spring moves with velocity $(y/l)\dot{x}$.

$T_{spring} = \int \tfrac{1}{2}\, (m_s/l)\, dy\, (y\,\dot{x}/l)^2 = \tfrac{1}{6}\, m_s\, \dot{x}^2 \rightarrow$ effective mass $= m + m_s/3$

$$\omega_n = \sqrt{(\, k \,/\, (m + m_s/3)\,)} \quad (5)$$

3. Equivalent Systems (Equivalent Stiffness)

Several springs can be replaced by one equivalent spring k_{eq} without changing the natural frequency.



Parallel: $k_{eq} = k_1 + k_2$ Series: $1/k_{eq} = 1/k_1 + 1/k_2$

Fig 1.3 Spring combinations

Elastic member (end mass)	Equivalent stiffness k
Cantilever, load at free end (length l)	$3EI / l^3$
Simply supported beam, load at centre	$48EI / l^3$
Fixed-fixed beam, load at centre	$192EI / l^3$
Axial rod (area A, length l)	AE / l
Torsion of shaft (length l)	$k_t = GJ / l$ (N·m/rad)

Beams in series with spring: add flexibilities (1/k). Beam and spring in parallel: add stiffnesses.

4. Torsional Vibration Systems

A disc of mass moment of inertia I ($kg·m^2$) on a shaft of length l, diameter d.

Angular displacement θ about the shaft axis.

Torsional stiffness $k_t = T/\theta = GJ/l$, polar moment $J = \pi d^4/32$.

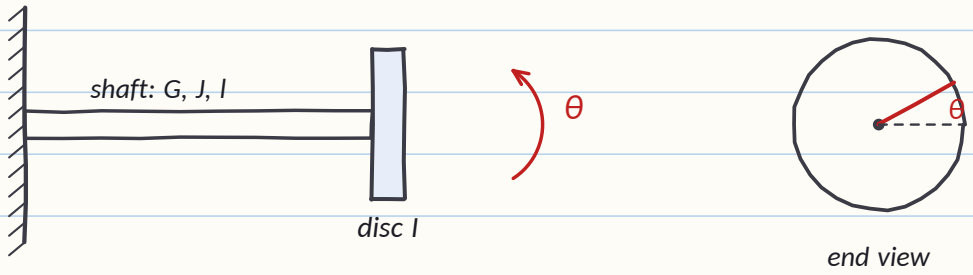


Fig 1.4 Torsional pendulum

Equation of motion: $I\ \ddot{\theta} + k_t\ \theta = 0$ ($\ddot{\theta}$ = angular acceleration)

$$\omega_n = \sqrt{(k_t/I)} = \sqrt{(\ GJ\ /\ (I\ l)\ )}\ rad/s \quad (6)$$

Two-rotor system

Two discs I_1, I_2 on a shaft, with torsional stiffness $k_t = GJ/l$ (between the discs). They vibrate in opposite directions; a node (zero amplitude section) lies between them.

Node position (from disc 1): both sides must have the same frequency, $\omega_n^2 =$

$GJ/(l_1 l_1) = GJ/(l_2 l_2)$, with $l_1 + l_2 = l$.

$$l_1 = l\, I_2 / (I_1 + I_2) \quad l_2 = l\, I_1 / (I_1 + I_2)$$
$$\omega_n = \sqrt{(GJ\, (I_1 + I_2) / (I_1\, I_2\, l))} \qquad (7)$$

- Geared system: refer all inertias/stiffnesses to one shaft using gear ratio $n = N_2/N_1$. $I_{equiv} = n^2 I$ (referred to motor shaft); $k_{t,equiv} = n^2 k_t$.

5. Damped Free Vibration

Real systems lose energy (friction, air, internal). Damping force opposes velocity.

Types: viscous, Coulomb (dry friction), hysteretic (structural).

5.1 Viscous damping ($F_d = c\,\dot{x}$)

Damper force proportional to velocity. Newton: $m\,\ddot{x} = - k\,x - c\,\dot{x}$

$$m\,\ddot{x} + c\,\dot{x} + k\,x = 0 \rightarrow \ddot{x} + 2\zeta\omega_n\,\dot{x} + \omega_n^2\,x = 0 \qquad (8)$$

Critical damping coefficient $c_c = 2\sqrt{(km)} = 2m\omega_n$ (dividing line between oscillatory & non-oscillatory motion). Damping ratio $\zeta = c/c_c$.

Characteristic eqn ($x = e^{st}$): $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$

$$s_{1,2} = - \zeta\omega_n \pm \omega_n \sqrt{(\zeta^2 - 1)} \qquad (9)$$

under damped ($\zeta < 1$)
critically damped ($\zeta = 1$)
over damped ($\zeta > 1$)

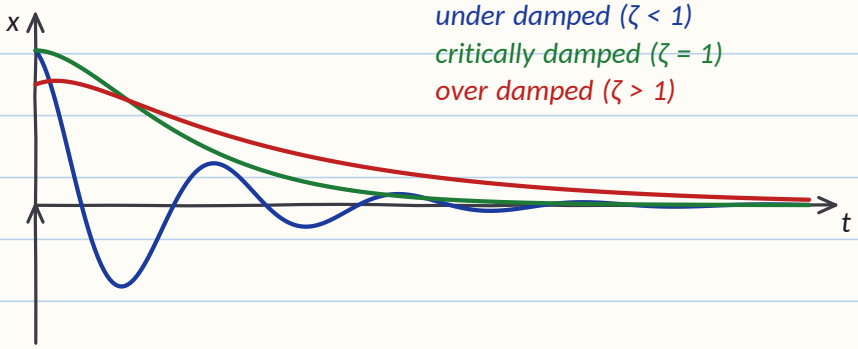


Fig 1.5 Response of viscously damped system

Case 1 : Under-damped ($\zeta < 1$, $c < c_c$)

Roots complex: $s = -\zeta\omega_n \pm i\,\omega_d$, $\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$ (damped natural frequency)

$$x(t) = e^{-\zeta\omega_n t} (A \cos \omega_d t + B \sin \omega_d t) = X\, e^{-\zeta\omega_n t} \cos(\omega_d t - \varphi)$$
$$A = x_0,\ B = (\dot{x}_0 + \zeta\omega_n x_0) / \omega_d \qquad (10)$$

- Oscillatory, amplitude decays exponentially (envelope $X\, e^{-\zeta\omega_n t}$). Time period $T_d = 2\pi/\omega_d$ (slightly more than undamped).

Case 2 : Critically damped ($\zeta = 1$, $c = c_c$)

Equal real roots $s = -\omega_n$. $x(t) = (A + B\,t)\,e^{-\omega_n t}$, $A = x_0$, $B = \dot{x}_0 + \omega_n x_0$

- No oscillation. Returns to equilibrium in minimum time. Used in guns recoil mechanism, door closers, instruments.

Case 3 : Over-damped ($\zeta > 1$, $c > c_c$)

Two distinct negative real roots $s_{1,2} = (-\zeta \pm \sqrt{(\zeta^2 - 1)})\,\omega_n$.

$x(t) = A\,e^{s_1 t} + B\,e^{s_2 t}$ - non-oscillatory, aperiodic, slow return to equilibrium.

Logarithmic decrement (δ)

Natural log of the ratio of any two successive amplitudes (one period T_d apart) of an under-damped system:

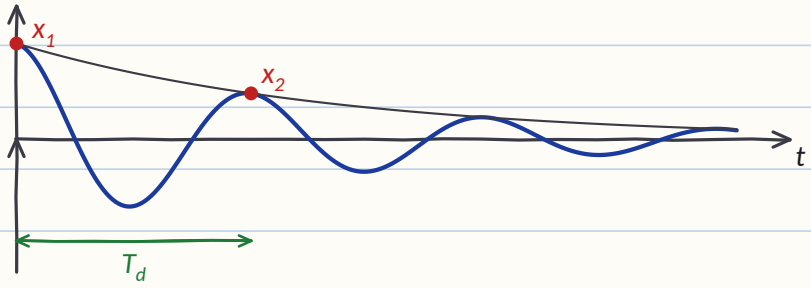


Fig 1.6 Successive amplitudes

$$\delta = \ln(x_1/x_2) = 2\pi\zeta / \sqrt{(1 - \zeta^2)} \approx 2\pi\zeta \text{ (for small } \zeta)$$

$$\text{after n cycles: } \delta = (1/n)\ln(x_0/x_n)\quad \zeta = \delta / \sqrt{(4\pi^2 + \delta^2)} \qquad (11)$$

Use:

From experiment: measure x_0 and x_n, find δ, then ζ and then $c = \zeta\, c_c$. This is the standard way of finding damping of a system.

5.2 Coulomb (dry-friction) damping

Friction force $F = \mu N = \mu mg$ (constant magnitude) always opposes the direction of motion. Mass sliding on rough surface with spring.

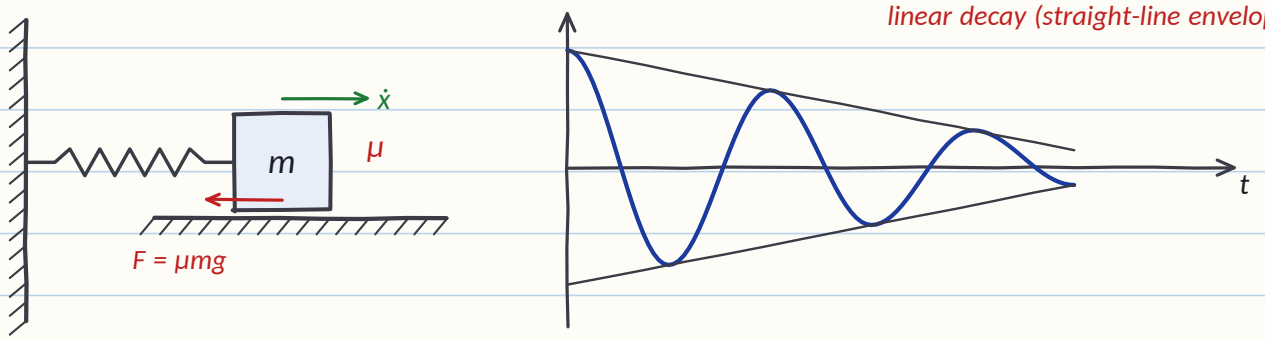


Fig 1.7 Coulomb damped system and response

- Equation: $m\ddot{x} + kx = -F$ when $\dot{x} > 0$ (moving right); $m\ddot{x} + kx = +F$ when $\dot{x} < 0$. Solution: $x = A\cos\omega_n t + B\sin\omega_n t \pm F/k$.
- Frequency is same as undamped: $\omega_n = \sqrt{(k/m)}$. The equilibrium position shifts to $\pm F/k$ alternately in each half cycle.
- Amplitude reduces by $2F/k$ per half cycle and $4F/k$ per full cycle (linear decay, not exponential).
- Motion stops when amplitude $\leq F/k$ (spring force cannot overcome friction). Final rest position may not be at $x = 0$.

$$x_n = x_0 - n\,(4F/k) \quad \text{No. of cycles} = (x_0 - F/k) / (4F/k)\ ,\ F = \mu m g \qquad (12)$$

5.3 Equivalent viscous damping coefficient (c_{eq})

To simplify, replace any non-viscous damping by an equivalent viscous damper that dissipates same energy per cycle at harmonic motion $x = X\sin\omega t$.

Viscous damper: $\Delta W = \int F_d\,dx = \int (c\dot{x})\dot{x}\,dt = \pi c \omega X^2$

Coulomb: $\Delta W = 4FX$ (force F over distance $4X$ per cycle). Equate: $\pi c_{eq}\omega X^2 = 4FX$

$$\text{Coulomb: } c_{eq} = 4F / (\pi\omega X)$$
$$\text{Quadratic } (F = c'\dot{x}^2)\text{: } c_{eq} = 8c'\omega X / (3\pi) \qquad (13)$$
$$\text{Hysteretic } (\Delta W = \pi h X^2)\text{: } c_{eq} = h/\omega$$

- Note: for Coulomb damping c_{eq} depends on X and ω - so it is amplitude dependent.

6. Solved Problems

Problem 1 (natural frequency from static deflection)

A mass of 10 kg suspended by a spring deflects 2 mm statically. Find k, ω_n, f_n.

$k = mg/\delta_{st} = 10(9.81)/0.002 = 49\,050$ N/m

$\omega_n = \sqrt{(g/\delta_{st})} = \sqrt{(9.81/0.002)} = 70.04$ rad/s; $f_n = \omega_n/2\pi = 11.15$ Hz

Problem 2 (energy method - uniform rod)

A uniform rod, mass m, length l, pivoted at one end swings in vertical plane. Find ω_n for small oscillations.

$I_O = ml^2/3$. $T = \tfrac{1}{2} I_O \dot{\theta}^2 = \tfrac{1}{6} m l^2 \dot{\theta}^2$. $U = mg(l/2)(1 - \cos\theta) \approx \tfrac{1}{4} mgl\theta^2$

$T_{max} = U_{max}$: $\tfrac{1}{2}(ml^2/3)\omega_n^2 = \tfrac{1}{2}(mgl/2) \rightarrow \omega_n = \sqrt{(3g/2l)}$.

For $l = 1$ m: $\omega_n = \sqrt{14.715} = 3.836$ rad/s, $T = 2\pi/\omega_n = 1.64$ s.

Problem 3 (cantilever with end mass)

Steel cantilever $E = 200$ GPa, $l = 0.4$ m, section 20 mm × 10 mm (thickness 10 mm) carries 5 kg at the free end. Find f_n. Neglect beam mass.

$I = b\,h^3/12 = 0.02(0.01)^3/12 = 1.667\times10^{-9}\ m^4$

$k = 3EI/l^3 = 3(200\times10^9)(1.667\times10^{-9})/(0.4)^3 =$ 15 625 N/m

$\omega_n = \sqrt{(15625/5)} = 55.9$ rad/s; $f_n =$ 8.90 Hz

Problem 4 (torsional system)

A disc of $I = 0.1$ kg·m^2 is fixed to a steel shaft ($G = 80$ GPa), $d = 20$ mm, $l = 0.5$ m, other end fixed. Find f_n.

$J = \pi d^4/32 = 1.571\times10^{-8}\ m^4$; $k_t = GJ/l = 80\times10^9(1.571\times10^{-8})/0.5 = 2513$ N·m/rad

$\omega_n = \sqrt{(k_t/I)} = \sqrt{(2513/0.1)} = 158.5$ rad/s; $f_n =$ 25.2 Hz

Problem 5 (viscous damping)

$m = 10$ kg, $k = 4000$ N/m, $c = 40$ N·s/m. Find ζ, ω_d, δ and ratio of successive amplitudes.

$\omega_n = \sqrt{(4000/10)} = 20$ rad/s; $c_c = 2\sqrt{(km)} = 2\sqrt{(40000)} = 400$ N·s/m; $\zeta = 40/400 =$ 0.1 (under-damped)

$\omega_d = 20\sqrt{(1 - 0.01)} =$ 19.90 rad/s; $\delta = 2\pi(0.1)/\sqrt{0.99} =$ 0.6315

$x_1/x_2 = e^{\delta} = e^{0.6315} =$ 1.88 (each amplitude is about 53% of the previous one).

Problem 6 (finding ζ from decay)

Vibration amplitude of a system falls to 25% of initial value after 5 cycles. Find ζ.

$\delta = (1/5)\ln(1/0.25) = 0.2\ (1.386) = 0.2773$

$\zeta = \delta / \sqrt{(4\pi^2 + \delta^2)} = 0.2773 / \sqrt{(39.48 + 0.077)} =$ 0.0441 (≈ 4.4 % of critical).

Problem 7 (Coulomb damping)

$m = 2$ kg, $k = 800$ N/m, $\mu = 0.15$. Mass pulled 50 mm and released. Find (i) amplitude loss per cycle, (ii) when the mass stops, (iii) c_{eq} for first cycle.

$F = \mu mg = 0.15(2)(9.81) = 2.943$ N; $F/k = 3.68$ mm; $\omega_n = \sqrt{(800/2)} = 20$ rad/s

(i) Loss per cycle $= 4F/k =$ 14.72 mm (7.36 mm per half cycle).

(ii) Amplitudes after half cycles: 42.6, 35.3, 27.9, 20.6, 13.2, 5.85 (> 3.68, still moves), then -1.51 mm (< F/k) → stops after 7 half cycles (3.5 cycles), 1.51 mm from the mean position. Time $= 7\pi/\omega_n = 1.10$ s.

(iii) $c_{eq} = 4F/(\pi\ \omega\ X) = 4(2.943)/(\pi \times 20 \times 0.05) =$ 3.75 N·s/m.

Problem 9 (equivalent stiffness)

Springs $k_1 = 3000$ and $k_2 = 6000$ N/m are in series; this combination is in parallel with $k_3 = 2000$ N/m. A 10 kg mass is attached. Find f_n.

Series: $k_{12} = k_1k_2/(k_1+k_2) = 18\times10^6/9000 = 2000$ N/m. Parallel: $k_{eq} = 2000 + 2000 =$ 4000 N/m

$\omega_n = \sqrt{(4000/10)} = 20$ rad/s; $f_n = 3.18$ Hz.

Problem 10 (U-tube manometer - energy method)

Liquid column of total length L in a U-tube of area A, density ρ. Find ω_n.

Displace the level by x: mass $m = \rho AL$; $T = \tfrac{1}{2}\,\rho AL\,\dot{x}^2$. Net head difference 2x so $U = \tfrac{1}{2}\,(2\rho gA)\,x^2$.

$T_{max} = U_{max} \rightarrow \omega_n^2 = 2\rho gA / \rho AL \rightarrow \omega_n = \sqrt{(2g/L)}$. For L = 0.5 m: $\omega_n = 6.26$ rad/s, $f_n =$ 1.0 Hz.

Problem 11 (two-rotor torsional system)

Two rotors $I_1 = 0.8$ and $I_2 = 0.4$ kg·m² on a steel shaft d = 40 mm, l = 1 m, G = 80 GPa. Find f_n and node position.

$J = \pi(0.04)^4/32 = 2.513\times10^{-7}$ m⁴; GJ = 20 106 N·m²

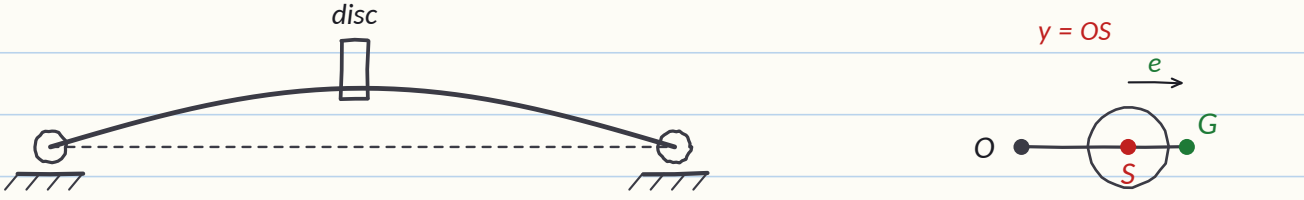
$\omega_n = \sqrt{(\ GJ\ (I_1+I_2)\ /\ (I_1I_2l)\ )} = \sqrt{(20106\times1.2/0.32)} = 274.6$ rad/s, $f_n = 43.7$ Hz

Node: $l_1 = l\ I_2/(I_1+I_2) = 0.4/1.2 = 0.333$ m from the 0.8 kg·m² rotor ($I_2 = 0.667$ m).

7. Whirling of Shafts

Whirling (whipping) : at certain speeds a rotating shaft bows and the bent shaft itself rotates (whirls) in a plane about the bearing axis. Causes: mass eccentricity, non-uniform shaft, unbalance, gyroscopic effects, bearing misalignment.

Critical speed : speed at which whirl amplitude becomes dangerously large (resonance). It equals the natural frequency of transverse vibration of the shaft.



bearings & deflected shaft — O: bearing axis, S: shaft centre, G: C.G.

Fig 1.8 Whirling of a shaft with a disc

Critical speed - derivation (no damping)

Disc mass m, eccentricity e (SG), shaft stiffness k (transverse), shaft deflection y (OS), shaft speed ω.

- Spring (elastic) force on disc, towards O: $k\,y$.
- Centrifugal force on disc (C.G. at distance $y + e$ from O): $m\,\omega^2\,(y + e)$.

Balance: $k\,y = m\,\omega^2\,(y + e) \rightarrow y\,(k - m\omega^2) = m\,\omega^2\,e$

$$y = e\,r^2 / (1 - r^2),\ r = \omega / \omega_n,\ \omega_n = \sqrt{(k/m)} \quad (14)$$

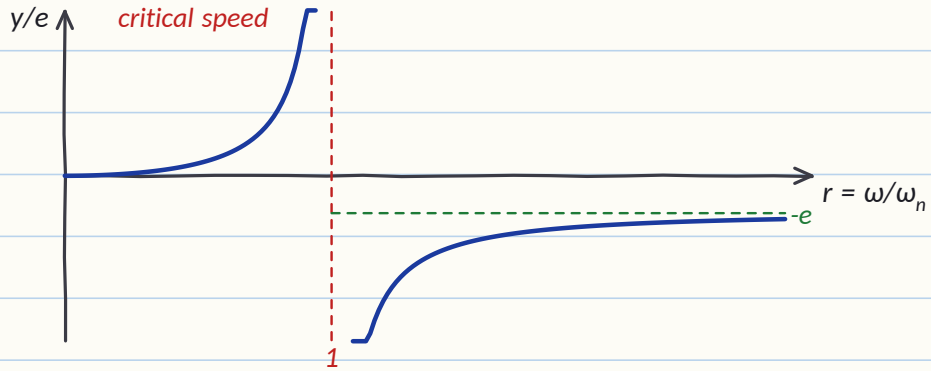


Fig 1.9 Whirl amplitude vs speed ratio

- $r < 1$: y positive; C.G. G lies outside S (heavy side out). y rises to ∞ as $r \rightarrow 1$ (resonance).
- $r = 1$: $\omega = \omega_n$ = critical speed: $\omega_c = \sqrt{(k/m)} = \sqrt{(g/\delta)}$ (δ = static deflection of shaft under disc).
- $r > 1$: y negative, $y \rightarrow -e$ as $r \rightarrow \infty$: G lies between O and S - shaft self-centres; running well above critical speed is smooth (turbines).

$$N_c = (60/2\pi)\,\sqrt{(g/\delta)}\ \text{rpm} \approx 946 / \sqrt{\delta}\ (\delta \text{ in mm}) \quad (15)$$

With damping: $y = e\,r^2 / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$ - finite at resonance.

Dunkerley's approximation (lower bound)

Shaft carrying several discs: exact fundamental critical speed is difficult; Dunkerley's empirical formula is simple, gives a lower bound (slightly lower than true value).

$$1/\omega_c^2 = 1/\omega_1^2 + 1/\omega_2^2 + 1/\omega_3^2 + \ldots + 1/\omega_s^2$$
$$\text{with } \omega_i^2 = g / \delta_i\ (\delta_i = \text{deflection under load i acting alone}) \quad (16)$$

- ω_i : critical speed if only the i-th disc acted on the shaft. ω_s : critical speed of shaft alone (its own mass); often ignored if shaft is light.
- Equivalent form: $1/\omega_c^2 = (\delta_1 + \delta_2 + \delta_3 + \ldots)/g$.

Deflection of simply supported shaft (span l) under load W at distance a from one

end, $b = l - a$: $\delta = W a^2 b^2 / (3 E I l)$; at centre $\delta = W l^3 / (48 E I)$.

Problem 8 (Dunkerley)

Shaft $d = 20$ mm, $l = 1$ m, $E = 200$ GPa, simply supported, carries 10 kg at 0.25 m from left bearing and 20 kg at mid-span. Ignoring shaft mass, find the critical speed.

$I = \pi d^4/64 = 7.854\times10^{-9}\ m^4$; $EI = 1570.8\ N\cdot m^2$

$\delta_1 = W_1 a^2 b^2/(3EIl) = 98.1(0.25)^2(0.75)^2/(3\times1570.8\times1) = 0.732$ mm

$\delta_2 = W_2 l^3/(48EI) = 196.2/(48\times1570.8) = 2.602$ mm

$\omega_1^2 = g/\delta_1 = 13\,403$; $\omega_2^2 = g/\delta_2 = 3770$; $1/\omega_c^2 = 7.46\times10^{-5} + 2.65\times10^{-4} = 3.40\times10^{-4}$

$\omega_c = 54.3$ rad/s $\rightarrow N_c = 60\omega_c/2\pi = \approx 518$ rpm.

Problem 12 (critical speed and whirl amplitude)

A 12 mm steel shaft ($E = 200$ GPa), 0.6 m between simple bearings, carries an 8 kg disc at the centre. Find critical speed, and whirl amplitude at 900 rpm if $e = 0.5$ mm.

$I = \pi(0.012)^4/64 = 1.018\times10^{-9}\ m^4$; $\delta = Wl^3/(48EI) =$

$78.48(0.216)/(48\times200\times10^9\times1.018\times10^{-9}) = 1.735$ mm

$\omega_c = \sqrt{(g/\delta)} = \sqrt{(9.81/1.735\times10^{-3})} = 75.2$ rad/s $\rightarrow N_c \approx 718$ rpm

$r = 900/718 = 1.2535$; $y = e\, r^2/(1 - r^2) = 0.5(1.571)/(1 - 1.571) = -1.375$ mm

(magnitude 1.375 mm; negative → above critical, G between O and S).

8. Quick Revision Table

Quantity	Formula
Natural frequency	$\omega_n = \sqrt{(k/m)} = \sqrt{(g/\delta_{st})}$; torsional $\sqrt{(k_t/I)}$
Damping ratio / critical	$\zeta = c/c_c$; $c_c = 2\sqrt{(km)} = 2m\omega_n$
Damped frequency	$\omega_d = \omega_n\sqrt{(1 - \zeta^2)}$
Log decrement	$\delta = \ln(x_1/x_2) = 2\pi\zeta/\sqrt{(1 - \zeta^2)} \approx 2\pi\zeta$
Coulomb decay / cycle	$4F/k$, $F = \mu mg$; stops if amplitude $\le F/k$
Equivalent damping	Coulomb: $4F/(\pi\omega X)$; quadratic: $8c'\omega X/(3\pi)$
Whirl amplitude	$y = e\, r^2/(1 - r^2)$; $\omega_c = \sqrt{(g/\delta)}$
Dunkerley	$1/\omega_c^2 = \Sigma\ 1/\omega_i^2$ (lower bound)

9. Short Questions & Answers

1. **Define degree of freedom.** Minimum number of independent co-ordinates needed to define the configuration of the system.
2. **Why is the natural frequency independent of amplitude?** In a linear system the restoring force is ~ displacement, so $\omega_n = \sqrt{(k/m)}$ depends only on k and m.
3. **What is critical damping?** The minimum damping ($c = 2\sqrt{(km)}$) at which the system does not oscillate; returns to rest in least time.
4. **Define logarithmic decrement.** Natural log of ratio of two successive amplitudes, $\delta = \ln(x_n/x_{n+1}) = 2\pi\zeta/\sqrt{(1-\zeta^2)}$.
5. **How does Coulomb damping differ from viscous?** Force is constant (μN), decay is linear, motion stops in finite time, frequency unchanged; viscous gives exponential decay, changes frequency to ω_d.
6. **What is meant by equivalent viscous damping?** A viscous damper that dissipates the same energy per cycle as the actual (non-viscous) damping.
7. **What is critical speed of a shaft?** Speed at which the rotational speed equals the natural frequency of transverse vibration; whirl amplitude becomes very large.
8. **Why does Dunkerley's method give a lower bound?** It neglects the interaction of loads and adds flexibilities; the result is always slightly below the true fundamental frequency.
9. **Why is it safe to run above critical speed?** For $r > 1$ the shaft self-centres ($y \to -e$); the speed must be passed through quickly to avoid build-up of amplitude.
10. **Effect of spring mass on ω_n?** One third of spring mass is added to the attached mass: $m_{eff} = m + m_s/3$, so ω_n reduces.

Important questions: (1) derive ω_n by energy method (2) three damping cases with sketches (3) log decrement derivation (4) Coulomb damping decay & c_{eq} (5) critical speed of shaft with a disc (6) Dunkerley's method problem.

UNIT - 2

Forced Vibrations of SDOF Systems

Subject: Mechanical Vibrations | Code: 23A03702a

Syllabus: Steady state forced vibration, sources of excitation, impressed harmonic force, resonance, impressed force due to unbalance, motion excitation, transmissibility and isolation, performance of different types of isolators, power absorbed by viscous damping.

1. Forced Vibration - Introduction

Forced vibration : the system is subjected to an external force (or motion) that acts continuously. Response has two parts:

- *Transient (complementary function, free vibration) - dies out due to damping.*
- *Steady state (particular integral) - persists at the frequency of excitation ω, whatever the natural frequency. This is what we study.*

Sources of excitation

Source	Examples
Unbalance (rotating mass)	Turbines, motors, fans, flywheels, wheels of vehicles - force $m_0 e\omega^2$
Reciprocating motion	IC engines, compressors, pumps - inertia forces of piston/crank
Motion of support (base)	Road roughness, earthquake, vibrating foundation, ship deck
Impact / periodic force	Forging hammers, press, gear tooth errors, cam-follower, blade passing
Fluid flow / wind	Vortex shedding, flutter, wave loading, pulsating pressure
Misalignment, loose parts	Couplings, shaft bend, bearing defects

2. Impressed Harmonic Force (SDOF with damping)

Mass m, spring k, damper c, acted by $F(t) = F_0 \sin \omega t$ (F_0 = force amplitude, ω = forcing frequency).

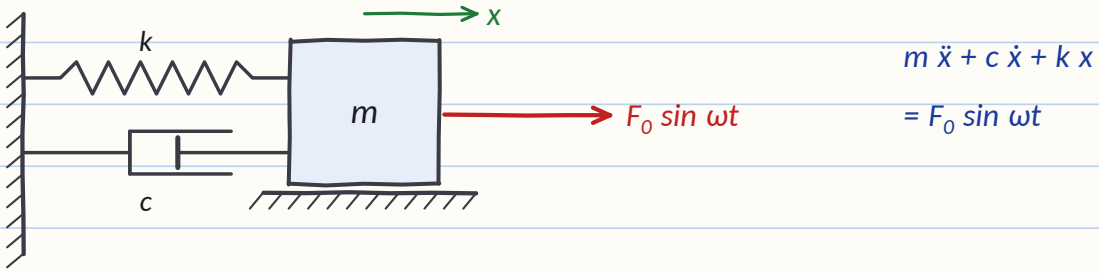


Fig 2.1 SDOF system with harmonic force

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega t \quad (1)$$

Steady state solution (same frequency, lagging by phase φ): $x(t) = X \sin(\omega t - \varphi)$.

Substitute in (1) and equate the sine and cosine terms:

$$X = F_0 / \sqrt{[\,(k - m\omega^2)^2 + (c\omega)^2\,]} \qquad \varphi = \tan^{-1}[\,c\omega / (k - m\omega^2)\,] \quad (2)$$

Divide by k, use $r = \omega/\omega_n$, $\zeta = c/c_c$, static deflection $X_{st} = F_0/k$:

$$M.F. = X / X_{st} = 1 / \sqrt{[\,(1 - r^2)^2 + (2\zeta r)^2\,]}$$

$$\varphi = \tan^{-1}[\,2\zeta r / (1 - r^2)\,] \quad (0 \le \varphi \le 180°) \quad (3)$$

M.F. = magnification factor (dynamic amplification, also called dynamic magnifier).

Complete solution = transient + steady state

General solution of (1): $x(t) = x_c(t) + x_p(t)$

$$x(t) = e^{-\zeta \omega_n t}(\,A \cos \omega_d t + B \sin \omega_d t\,) + X \sin(\omega t - \varphi) \quad (2a)$$

- First term (transient) vanishes after a few time constants $1/(\zeta\omega_n)$; A and B come from initial conditions.
- Only the second term (steady state) is important for design - amplitude is independent of initial conditions.

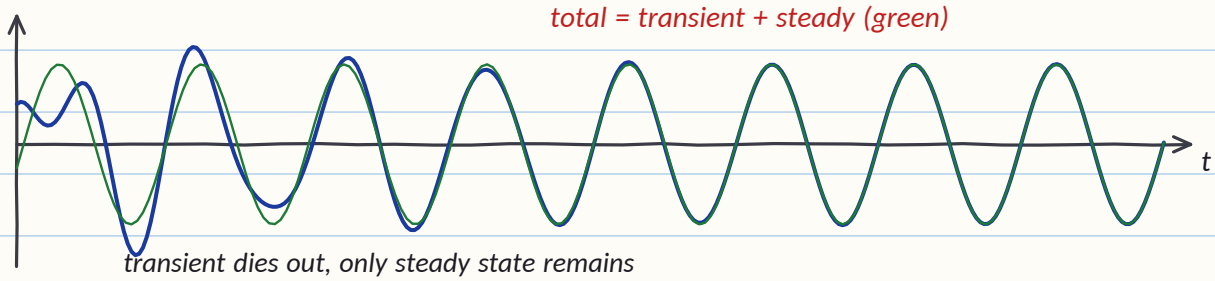


Fig 2.1a Build-up of forced response

Complex (phasor) method - frequency response function

Write $F = F_0 e^{i\omega t}$, $x = X^* e^{i\omega t}$ (X^* complex, includes phase). Then $(-m\omega^2 + ic\omega + k)\,X^* = F_0$ →

$$H(\omega) = X^*/F_0 = 1 / (k - m\omega^2 + ic\omega) \quad |H| = 1/\sqrt{[(k - m\omega^2)^2 + (c\omega)^2]}$$

$$\text{dimensionless: } H(r)\ k = 1 / (1 - r^2 + i\,2\zeta r) \qquad \textbf{(2b)}$$

The modulus gives amplitude ratio, the argument gives phase lag $\varphi = \tan^{-1}[2\zeta r/(1 - r^2)]$. This is how frequency response is measured experimentally.

Undamped forced vibration ($\zeta = 0$)

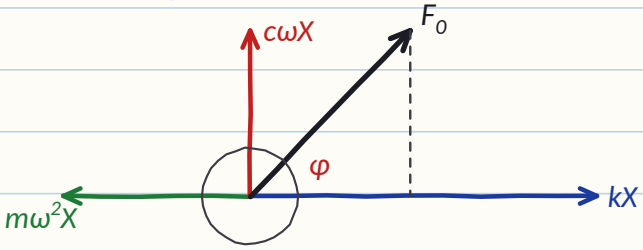
$m\ddot{x} + k\,x = F_0 \sin \omega t$. Particular solution $X = F_0/(k - m\omega^2) = X_{st}/(1 - r^2)$; response in phase ($r < 1$) or 180° out of phase ($r > 1$).

Complete solution with $x_0 = \dot{x}_0 = 0$: $x = X_{st}/(1 - r^2) \cdot (\sin \omega t - r \sin \omega_n t)$

- Beating: when ω is close to ω_n, the two frequencies add to give a slowly varying amplitude (beat frequency = $\omega_n - \omega$).
- At resonance ($\omega = \omega_n$): $x = (X_{st}/2)(\sin \omega_n t - \omega_n t \cos \omega_n t) \rightarrow$ amplitude grows linearly with time ($\sim t$); failure is a matter of time.

Vector (force polygon) representation

Four rotating vectors in equilibrium: spring force kX, damping force $c\omega X$ (leads by 90° over spring), inertia force $m\omega^2 X$ (opposite to spring), applied force F_0.



$F_0^2 = (k - m\omega^2)^2X^2 + (c\omega X)^2$

$\tan \varphi = c\omega / (k - m\omega^2)$

x lags F_0 by φ

Fig 2.2 Force polygon

3. Resonance and Frequency-Response Curves

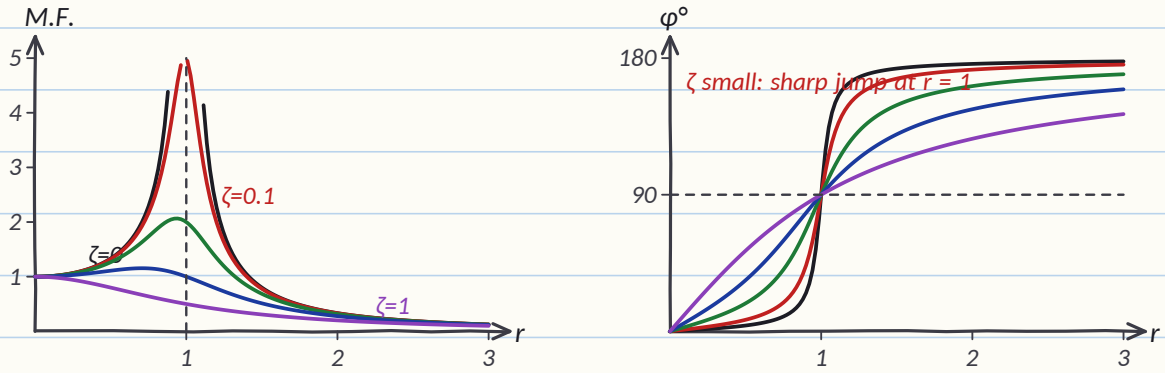


Fig 2.3 Magnification factor and phase angle vs frequency ratio r

Resonance : $r = 1$ ($\omega = \omega_n$). Amplitude is limited only by damping:

$$\text{At } r = 1: X = X_{st} / 2\zeta = F_0/(c\omega_n),\ \varphi = 90° \quad (4)$$

- Undamped ($\zeta = 0$) at resonance : $X \to \infty$ (grows linearly with time) - dangerous.
- Peak of M.F. occurs slightly left of $r = 1$: $r_{peak} = \sqrt{(1 - 2\zeta^2)}$; $M.F._{max} = 1 / (2\zeta\sqrt{(1 - \zeta^2)}) \approx 1/2\zeta$.
- No peak exists if $\zeta > 0.707$ ($1/\sqrt{2}$) - M.F. decreases monotonically from 1.

Peak amplitude - derivation

Let $D = (1 - r^2)^2 + (2\zeta r)^2$. Maximise M.F. $= D^{-1/2}$: $dD/dr = 0 \to -4r(1 - r^2) + 8\zeta^2 r = 0$

$$r_{peak}^2 = 1 - 2\zeta^2 \to r_{peak} = \sqrt{(1 - 2\zeta^2)}$$
$$M.F._{max} = 1 / (2\zeta \sqrt{(1 - \zeta^2)}) \text{ (substituting back)} \quad (4a)$$

- $r << 1$ ($\omega << \omega_n$): M.F. ≈ 1, $\varphi \approx 0$ - stiffness controlled, system follows force quasi-statically.
- $r = 1$: damping controlled ; force balances damper force; $\varphi = 90°$.
- $r >> 1$: M.F. $\approx 1/r^2 \to 0$, $\varphi \to 180°$ - mass controlled; mass hardly moves (basis of seismic instruments & isolators).

Bandwidth, half-power points and Q-factor

Frequencies ω_1, ω_2 where amplitude falls to (peak)/$\sqrt{2}$ (power falls to half). For small ζ:

$$\omega_{1,2} \approx \omega_n(1 \pm \zeta),\ Q = 1/2\zeta = \omega_n / (\omega_2 - \omega_1)$$
$$\zeta = (\omega_2 - \omega_1) / (2\omega_n) \text{ - used to measure damping experimentally} \quad (5)$$

Problem 1 (steady-state amplitude & phase)

$m = 10$ kg, $k = 4000$ N/m, $c = 40$ N·s/m, $F_0 = 100$ N at $\omega = 15$ rad/s. Find X and φ.

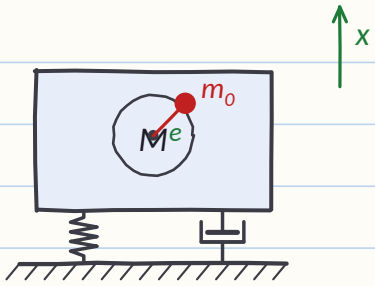
$\omega_n = 20$ rad/s, $\zeta = 40/400 = 0.1$, $r = 15/20 = 0.75$, $X_{st} = 100/4000 = 25$ mm

M.F. $= 1/\sqrt{[(1 - 0.5625)^2 + (2\times0.1\times0.75)^2]} = 1/\sqrt{(0.1914 + 0.0225)} = 2.162$

$X = 25 \times 2.162 = 54.05$ mm; $\varphi = \tan^{-1}[0.15/0.4375] = 18.9°$

4. Excitation by Rotating Unbalance

Machine total mass m (including eccentric mass m_0) on spring-damper, constrained to move vertically. Eccentric mass m_0 at eccentricity e rotates at ω.



Vertical component of centrifugal force

$F = m_0\, e\, \omega^2 \sin \omega t$

$m = M$ (total mass)

Fig 2.4 Rotating unbalance excitation

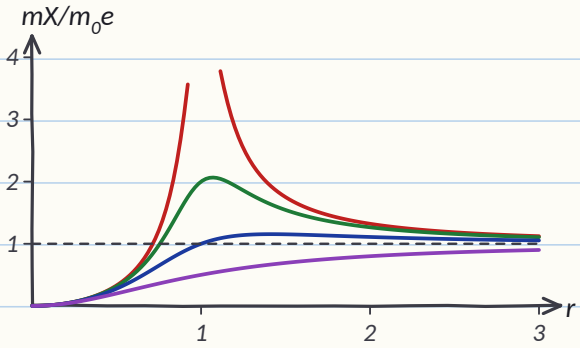
Centrifugal force vertical component $m_0\, e\, \omega^2 \sin \omega t$ - force amplitude depends on ω^2 (grows with speed).

$$m\ddot{x} + c\dot{x} + kx = m_0\, e\, \omega^2 \sin \omega t \quad (6)$$

Same as (1) with $F_0 \to m_0 e \omega^2$:

$$X = m_0\, e\, \omega^2 / \sqrt{[(k - m\omega^2)^2 + (c\omega)^2]}$$

$$m\,X / (m_0\, e) = r^2 / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]} \quad \varphi = \tan^{-1}[2\zeta r/(1 - r^2)] \quad (7)$$



At $r = 1$: $mX/(m_0 e) = 1/2\zeta$

$r \to \infty$: $X \to m_0 e / m$

(independent of damping)

$r \to 0$: $X \to 0$

ζ=0.1 ζ=0.25 ζ=0.5 ζ=1

Fig 2.5 Response to rotating unbalance

- *Peak occurs at $r = 1/\sqrt{(1 - 2\zeta^2)}$ (slightly right of resonance); $M.F._{max} = 1 / (2\zeta\sqrt{(1 - \zeta^2)})$.*
- *Reciprocating machines: the primary inertia force $\approx m_0 e\omega^2 \cos \omega t$ - the same analysis applies (m_0 = reciprocating mass).*
- *Force transmitted to foundation: $F_T = X\sqrt{(k^2 + (c\omega)^2)} = kX\sqrt{(1 + (2\zeta r)^2)}$.*

Problem 2 (rotating unbalance)

Machine $M = 200$ kg on springs $k = 80$ kN/m, $c = 800$ N·s/m; unbalance $m_0 e = 0.4$ kg·m; speed 300 rpm. Find amplitude, phase and amplitude at resonance.

$\omega_n = \sqrt{(80000/200)} = 20$ rad/s; $\zeta = 800/(2\sqrt{(80000 \times 200)}) = 800/8000 = 0.1$; $\omega = 2\pi(300)/60 = 31.42$ rad/s; $r = 1.571$

$mX/(m_0 e) = r^2/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]} = 2.467/\sqrt{(2.153 + 0.0987)} = 2.467/1.501 = 1.644$

$X = 1.644 \times 0.4/200 = 3.29$ mm; $\varphi = 180° - \tan^{-1}(0.314/1.467) = 167.9°$

At resonance (≈ 191 rpm): $X = (m_0 e/m)/(2\zeta) = 0.002/0.2 = 10$ mm.

Problem 2a (limiting the amplitude)

For the machine of Problem 2 find the speed at which amplitude is maximum, and the maximum amplitude.

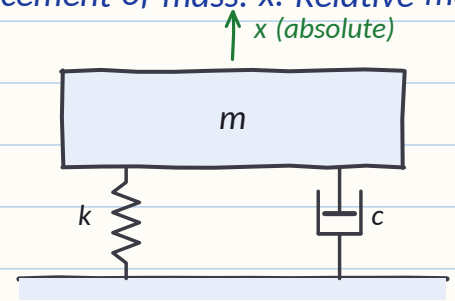
$r_{peak} = 1/\sqrt{(1 - 2\zeta^2)} = 1/\sqrt{0.98} = 1.010 \rightarrow \omega = 1.010 \times 20 = 20.2$ rad/s $\rightarrow N = 193$ rpm

$(mX/m_0 e)_{max} = 1/(2\zeta\sqrt{(1 - \zeta^2)}) = 1/(0.2 \times 0.995) = 5.025 \rightarrow X_{max} = 5.025 \times 0.002 = 10.05$ mm

Well above resonance ($r \geq 3$, i.e. $N > 570$ rpm) the amplitude approaches $m_0 e/m = 2$ mm ($r = 3$ gives 2.24 mm) - run the machine well above resonance.

5. Motion Excitation (Support / Base Excitation)

Base moves harmonically $y = Y \sin \omega t$ (road, foundation, earthquake). Absolute displacement of mass: x. Relative motion: $z = x - y$.



Eqn: $m\ddot{x} + c(\dot{x} - \dot{y}) + k(x - y) = 0$

or $m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky$

RHS = $Y\sqrt{(k^2 + (c\omega)^2)}\ \sin(\omega t + \alpha)$

Fig 2.6 Base-excited system

Solution $x = X \sin(\omega t - \varphi_1)$. Ratio of mass amplitude to base amplitude (displacement transmissibility):

$$X / Y = \sqrt{[1 + (2\zeta r)^2]} / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$$

$$\varphi_1 = \tan^{-1}[\ 2\zeta r^3 / (1 + (4\zeta^2 - 1)\ r^2)\ ] \quad (8)$$

Relative motion $z = x - y$ (what a vibrometer / seismic instrument measures):

$$m\ddot{z} + c\dot{z} + kz = m\omega^2 Y \sin \omega t \rightarrow Z / Y = r^2 / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]} \quad (9)$$

- Force transmitted to base through spring + damper: $F_T = k(x-y) + c(\dot{x}-\dot{y}) = -m\ddot{x} \rightarrow$ amplitude $F_T = m\omega^2 X$.

Problem 3 (base excitation)

$m = 50$ kg, $k = 20$ kN/m, $\zeta = 0.2$; base moves with $Y = 5$ mm at $\omega = 30$ rad/s. Find X and force transmitted.

$\omega_n = \sqrt{(20000/50)} = 20$ rad/s; $r = 1.5$; $2\zeta r = 0.6$

$X/Y = \sqrt{(1 + 0.36)}/\sqrt{[(1 - 2.25)^2 + 0.36]} = 1.166/1.387 = 0.841 \rightarrow X = 4.21$ mm

$F_T = m\,\omega^2 X = 50\,(900)(0.00421) = 189$ N.

Problem 4 (vehicle on a wavy road)

A car of 1200 kg, suspension $k = 400$ kN/m, $\zeta = 0.5$, moves at 100 km/h on a road with sinusoidal profile, amplitude 50 mm, wavelength 6 m. Find the amplitude of vertical motion of the car body and relative to the wheels.

$v = 27.78$ m/s; $\omega = 2\pi v/\lambda = 2\pi(27.78)/6 = 29.09$ rad/s; $\omega_n = \sqrt{(400000/1200)} = 18.26$ rad/s; $r = 1.593$

$2\zeta r = 1.593$; $X/Y = \sqrt{(1 + 2.539)}/\sqrt{[(1 - 2.539)^2 + 2.539]} = 1.881/2.215 = 0.849$

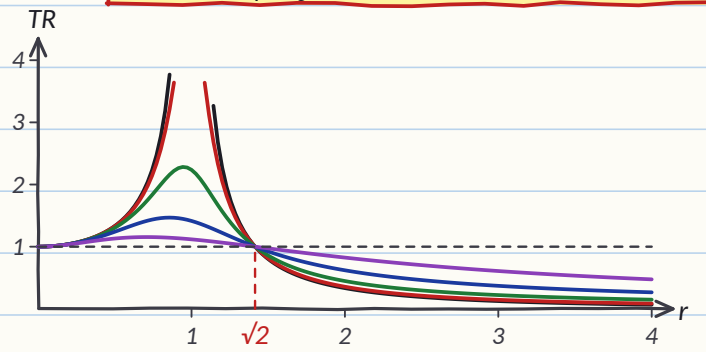
$X = 0.849 \times 50 = 42.5$ mm. Relative: $Z/Y = r^2/\sqrt{[\ldots]} = 2.539/2.215 = 1.146 \rightarrow Z = 57.3$ mm (suspension travel).

6. Transmissibility and Vibration Isolation

Vibration isolation : reducing transmission of vibration from machine to foundation (force isolation) or from foundation to equipment (motion isolation). Done by placing resilient mounts (isolators).

Transmissibility (TR) = ratio of transmitted force amplitude to applied force amplitude (or output motion to base motion). For both cases:

$$TR = F_T/F_0 = X/Y = \sqrt{[1 + (2\zeta r)^2]} \,/\, \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]} \qquad (10)$$



$r < \sqrt{2}$: amplification

$r > \sqrt{2}$: isolation ($TR < 1$)

all curves cross at $r = \sqrt{2}$

ζ=0 ζ=0.1 ζ=0.25

ζ=0.5 ζ=1

Fig 2.7 Transmissibility curves

- Isolation ($TR < 1$) only when $r > \sqrt{2}$, i.e. $\omega_n < \omega/\sqrt{2}$. So the isolator must be soft (low $k \rightarrow$ large static deflection).
- For $r > \sqrt{2}$, more damping increases TR (worse isolation); for $r < \sqrt{2}$ damping helps (limits resonance). Compromise: small damping ($\zeta \approx 0.05$-0.2) to pass resonance safely at start-up.

- Undamped: $TR = 1/(r^2 - 1)$ (magnitude). Isolation efficiency $\varepsilon = (1 - TR) \times 100\ \%$.

$$r = \sqrt{(1 + 1/TR)}\ \text{(undamped)};\ \delta_{st} = g/\omega_n^2;\ f_n = 15.76/\sqrt{\delta_{st}}(\text{mm})\ \text{Hz} \quad (11)$$

Problem 5 (design of isolator)

A 500 kg machine runs at 1200 rpm. Required isolation 80% (TR = 0.2); neglect damping. Find k and static deflection.

$\omega = 2\pi(1200)/60 = 125.7$ rad/s; $r = \sqrt{(1 + 1/0.2)} = \sqrt{6} = 2.449$; $\omega_n = 125.7/2.449 = 51.3$ rad/s

$k = m\,\omega_n^2 = 500(51.3)^2 = 1.316 \times 10^6$ N/m; $\delta_{st} = g/\omega_n^2 = 9.81/2632 = 3.73$ mm

(Total k; with 4 mounts each k = 329 kN/m.)

Problem 6 (force transmitted with damping)

Machine 120 kg, unbalanced force 400 N at 1800 rpm, on mounts of total $k = 1.2 \times 10^5$ N/m, $\zeta = 0.1$. Find the amplitude and force transmitted.

$\omega = 188.5$ rad/s; $\omega_n = \sqrt{(120000/120)} = 31.62$ rad/s; $r = 5.96$; $2\zeta r = 1.192$

$M.F. = 1/\sqrt{[(1 - 35.53)^2 + 1.421]} = 1/34.55 = 0.0289$; $X_{st} = 400/120000 = 3.33$ mm

$X = 3.33 \times 0.0289 = 0.096$ mm; $TR = \sqrt{(1 + 1.421)} \times 0.0289 = 0.045 \rightarrow F_T = 0.045 \times 400 = 18$ N (95.5% isolation).

Isolation vs. absorption (comparison)

Isolator (this unit)	Dynamic vibration absorber (Unit III)
Soft mount between source and base	Auxiliary spring-mass attached to the machine
Works for a wide range of speeds ($r > \sqrt{2}$)	Works at one operating frequency (tuned)
Reduces transmitted force / motion	Reduces vibration of the main mass itself

7. Performance of Different Types of Isolators

Choice depends on load, frequency range, damping needed, environment (oil, heat), cost and space.

Isolator	Characteristics / Typical use
Steel coil springs	Large static deflection → low f_n; very low damping ($\zeta \approx 0.005$); durable; good low-frequency isolation (compressors, large machines). May transmit high-frequency noise ("surge"), so add rubber pad / damper.
Rubber / neoprene mounts	Moderate damping, used in shear or compression; compact, cheap, easy to mount (engine mounts, pumps). Deteriorates with oil, heat, ageing; limited deflection.
Cork pads	Used under heavy machinery and foundations; good for high frequency; small deflection → higher f_n; damping moderate; life limited by moisture.
Felt pads	Light loads, high-frequency noise isolation, cheap; creeps & compacts with time.
Air (pneumatic) springs	Very low f_n; stiffness adjustable by pressure; self-levelling; damping through orifice. Precision equipment; costly, needs air supply.
Wire-rope / metal mesh, hydraulic mounts	High damping with moderate stiffness; used where shock, wide temperature, or both isolation and damping are needed.

- *Effective isolator needs:* low natural frequency (large δ_{st}), enough damping for resonance crossing, stability against rocking, and compatibility with environment.
- *Foundation mass:* mounting machine on large inertia block lowers amplitude and keeps the C.G. low.

Design procedure for a passive isolator

1. Find the disturbing frequency ω (operating speed) and forcing amplitude.
2. Decide the required isolation / TR (e.g. 80%, 90%).
3. Find r from TR: $r = \sqrt{(1 + 1/TR)}$ (undamped) or from curves with the chosen ζ.
4. Natural frequency $\omega_n = \omega/r$; stiffness $k = m\omega_n^2$; $\delta_{st} = mg/k$.
5. Select isolator type to give k and a small ζ; check start-up resonance crossing and rocking stability.

- *Active / semi-active isolation:* sensors + actuators vary force or damping in real time (car magnetic-ride dampers, precision tables) - effective also at low frequency where passive mounts fail.

8. Power Absorbed by Viscous Damping

Damper force $F_d = c\,\dot{x}$. With $x = X\sin(\omega t - \varphi)$, velocity $\dot{x} = \omega X\cos(\omega t - \varphi)$. Energy dissipated per cycle:

$$\Delta W = \int_0^T c\,\dot{x}\cdot\dot{x}\,dt = \pi\, c\,\omega\, X^2 \quad (12)$$

Average power = $\Delta W / T$:

$$P_{avg} = \tfrac{1}{2}\, c\,\omega^2 X^2 = \tfrac{1}{2}\, F_0\,\omega\, X\sin\varphi \quad (13)$$

- Second form: power supplied by the exciting force = F_0 × (component of velocity in phase with F_0) - at steady state equals power dissipated in damper.
- Using X from (2): $P = \tfrac{1}{2}\, c\,\omega^2 F_0^2 / [(k - m\omega^2)^2 + (c\omega)^2]$. Maximum at resonance ($\omega = \omega_n$):

$$P_{max} = F_0^2 / (2c) \quad (14)$$

- Half-power points (ω_1, ω_2) are where $P = P_{max}/2$ - hence the name; ΔW is the area of hysteresis loop in force-displacement plot (ellipse).

Problem 7 (power absorbed)

For Problem 1 ($c = 40$, $\omega = 15$ rad/s, $X = 54.05$ mm): $P = \tfrac{1}{2}\, c\,\omega^2 X^2 = 0.5(40)(225)(0.05405)^2 = 13.1$ W.

Check: $\tfrac{1}{2}\, F_0\,\omega\, X\sin\varphi = 0.5(100)(15)(0.05405)(0.324) = 13.1$ W. Energy lost per cycle $\Delta W = \pi\, c\,\omega\, X^2 = \pi(40)(15)(0.05405)^2 = 5.5$ J ($= P \times T = 13.1 \times 0.419$).

Problem 8 (damping from half-power bandwidth)

Frequency-response test: resonance peak at 20 Hz; amplitude falls to 0.707 of peak at 18.8 Hz and 21.2 Hz. Find ζ and Q.

$\zeta = (f_2 - f_1)/(2f_n) = (21.2 - 18.8)/(2\times 20) = 0.06$; $Q = 1/2\zeta = 8.33$.

Problem 9 (damping from resonance amplitude)

A machine with static deflection $X_{st} = 1.2$ mm under F_0 shows amplitude 15 mm at resonance. Find ζ and c/c_c.

$X_{res} = X_{st}/2\zeta \rightarrow \zeta = X_{st}/(2X_{res}) = 1.2/30 = 0.04$ (4% of critical damping).

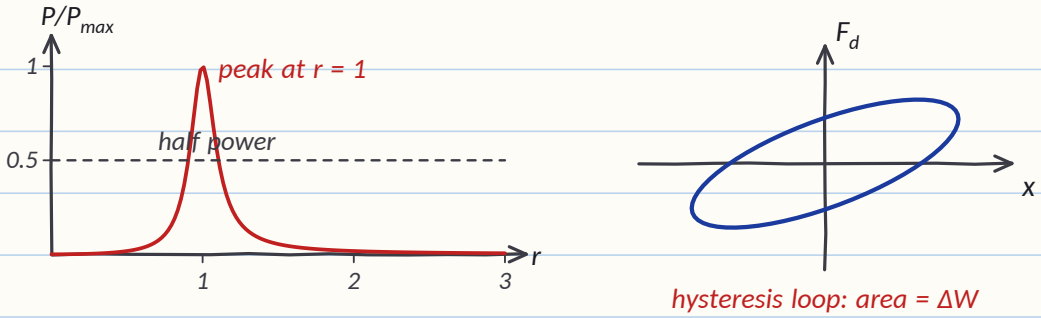


Fig 2.8 Power absorbed and damper energy loop

9. Quick Revision & Important Questions

Quantity	Formula
Force excitation	$M.F. = 1/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$; $\tan\varphi = 2\zeta r/(1 - r^2)$
Resonance	$r = 1$; $X = X_{st}/2\zeta$; $\varphi = 90°$
Unbalance	$mX/(m_0 e) = r^2 M.F.$; $X \to m_0 e/m$ as $r \to \infty$
Base excitation	$X/Y = \sqrt{(1 + (2\zeta r)^2)}$ M.F.; $Z/Y = r^2 M.F.$
Transmissibility	$TR = \sqrt{(1 + (2\zeta r)^2)}$ M.F.; isolation for $r > \sqrt{2}$
Power	$P = \frac{1}{2} c \omega^2 X^2$; $P_{max} = F_0^2/2c$

10. Short Questions & Answers

1. *What is resonance?* Condition $\omega = \omega_n$ ($r = 1$) where amplitude is maximum, limited only by damping; undamped amplitude $\to \infty$.
2. *What is magnification factor?* $X/X_{st} = 1/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$; ratio of dynamic to static amplitude.
3. *Why does the forced response occur at excitation frequency?* The transient dies out and the particular solution has the same frequency as the force.
4. *Define transmissibility.* Ratio of transmitted force (or motion) amplitude to the applied force (or base motion) amplitude.
5. *When is isolation effective?* Only when $r > \sqrt{2}$; then $TR < 1$. Lower the natural frequency (larger static deflection).
6. *Effect of damping beyond $r = \sqrt{2}$?* Damping increases transmissibility there, so damping should be minimal for pure isolation.
7. *Why does unbalance response tend to $m_0 e/m$ at high speed?* The mass rotates about its centre of gravity (self-centring) - amplitude independent of damping.

8. What is Q-factor? $Q = 1/2\zeta = \omega_n/(\omega_2 - \omega_1)$; measure of sharpness of resonance.
9. What phase angle at resonance? $\varphi = 90°$, independent of damping.
10. Average power absorbed by damper? $P = \frac{1}{2} c \omega^2 X^2$; at resonance $P_{max} = F_0^2/(2c)$.

Important questions: (1) derive X and φ for harmonic force; draw M.F. & phase curves (2) rotating-unbalance response (3) base excitation and transmissibility, why isolation needs $r > \sqrt{2}$ (4) isolator design problem (5) types and performance of isolators (6) power absorbed by viscous damper.

UNIT - 3

Two Degree Freedom Systems

Subject: Mechanical Vibrations | Code: 23A03702a

Syllabus: Formulation of equation of motion, natural frequencies and modes of vibration by classical method, coupled pendulum, forced vibration, dynamic vibration absorber.

1. Introduction to Two Degree Freedom Systems

A system needing two independent co-ordinates to describe its motion = 2-DOF system. Examples: two masses connected by springs, double pendulum, torsional system of two rotors, vehicle (bounce + pitch).

- Two equations of motion (coupled), two natural frequencies $\omega_1 < \omega_2$.
- For each natural frequency there is a definite amplitude ratio X_2/X_1 - called the principal (normal) mode of vibration.
- In a principal mode both masses move harmonically at the same frequency, in phase or 180° out of phase, passing through equilibrium together.
- Free vibration is the superposition of the two normal modes (depends on initial conditions).
- If excited at a natural frequency, the system vibrates in the corresponding mode only.
- Number of natural frequencies = number of DOF. Node = point of zero amplitude in a mode.

Types of coupling

- Static (elastic) coupling : stiffness matrix has off-diagonal term ($k_{12} \neq 0$) - displacement of one co-ordinate produces force on the other.
- Dynamic (mass / inertia) coupling : mass matrix has off-diagonal term ($m_{12} \neq 0$) - acceleration of one co-ordinate produces force on the other.
- Choice of co-ordinates changes coupling. Principal (normal) co-ordinates uncouple the equations completely (both M and K diagonal).

2. Formulation of Equations of Motion

Consider masses m_1, m_2, springs k_1 (wall - m_1), k_2 (between masses), k_3 (m_2 - wall).

Displacements x_1, x_2 from equilibrium.

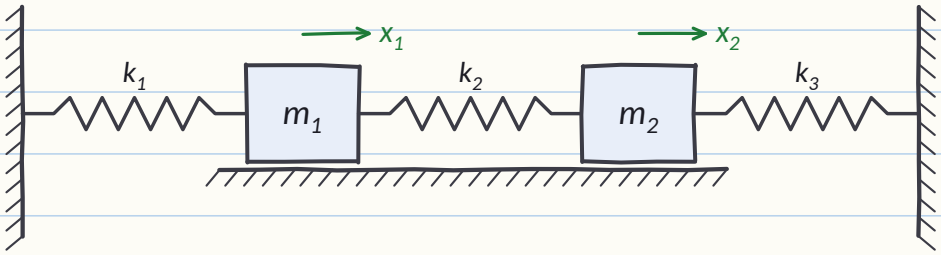


Fig 3.1 Two-DOF spring-mass system

Newton's second law for each mass:

$$m_1 \ddot{x}_1 = - k_1 x_1 + k_2(x_2 - x_1)$$
$$m_2 \ddot{x}_2 = - k_2(x_2 - x_1) - k_3 x_2 \quad (1)$$

Rearranged (coupled equations):

$$m_1 \ddot{x}_1 + (k_1 + k_2)\, x_1 - k_2\, x_2 = 0$$
$$m_2 \ddot{x}_2 - k_2\, x_1 + (k_2 + k_3)\, x_2 = 0 \quad (2)$$

Matrix form: $[M]\{\ddot{x}\} + [K]\{x\} = \{0\}$

$$[\, m_1\ 0\ ;\ 0\ m_2\,]\ \{\ddot{x}_1, \ddot{x}_2\} + [\, k_1{+}k_2\ -k_2\ ;\ -k_2\ k_2{+}k_3\,]\ \{x_1, x_2\} = \{0, 0\} \quad (3)$$

- $[M]$ diagonal (no dynamic coupling), $[K]$ symmetric with off-diagonal $-k_2$ (static coupling). Always $K_{ij} = K_{ji}$ (Maxwell reciprocity).
- With damping: $[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{F(t)\}$.

3. Natural Frequencies and Mode Shapes (Classical Method)

Assume harmonic solution, both masses at same frequency ω:

$$x_1 = X_1 \cos(\omega t + \varphi),\ x_2 = X_2 \cos(\omega t + \varphi) \quad (4)$$

Substitute in (2):

$$(k_1 + k_2 - m_1\omega^2)\, X_1 - k_2\, X_2 = 0$$
$$- k_2\, X_1 + (k_2 + k_3 - m_2\omega^2)\, X_2 = 0 \quad (5)$$

Non-trivial solution (X_1, X_2 ≠ 0) needs determinant of coefficients = 0 → frequency (characteristic) equation:

$$(k_1 + k_2 - m_1\omega^2)(k_2 + k_3 - m_2\omega^2) - k_2^2 = 0$$
$$m_1m_2\,\omega^4 - [\, m_1(k_2+k_3) + m_2(k_1+k_2)\,]\,\omega^2 + (k_1k_2 + k_2k_3 + k_3k_1) = 0 \quad (6)$$

Quadratic in ω^2: two positive real roots ω_1^2, ω_2^2 (natural frequencies; ω_1 is the fundamental).

Mode shapes (amplitude ratio) from eqn (5) for each root:

$$r_i = X_2/X_1\,|_{\omega\,=\,\omega i} = (k_1 + k_2 - m_1\omega_i^2) / k_2 = k_2 / (k_2 + k_3 - m_2\omega_i^2) \quad (7)$$

- Mode 1 (lower frequency): r_1 positive → masses move in phase, no node.
- Mode 2 (higher frequency): r_2 negative → masses move out of phase, one node between them.

General solution - superposition of the two modes (4 constants from initial conditions $x_1(0)$, $x_2(0)$, $\dot{x}_1(0)$, $\dot{x}_2(0)$):

$$x_1 = X_1^{(1)}\cos(\omega_1 t + \varphi_1) + X_1^{(2)}\cos(\omega_2 t + \varphi_2)$$
$$x_2 = r_1X_1^{(1)}\cos(\omega_1 t + \varphi_1) + r_2X_1^{(2)}\cos(\omega_2 t + \varphi_2) \quad (8)$$

Orthogonality of principal modes

Modal vectors $\{X\}^{(1)}$, $\{X\}^{(2)}$ satisfy: $\{X\}^{(1)T}\,[M]\,\{X\}^{(2)} = 0$ and $\{X\}^{(1)T}\,[K]\,\{X\}^{(2)} = 0$.

- Basis of modal analysis: with $x = [\Phi]\,q$, the equations decouple into two SDOF equations $M_i\,\ddot{q}_i + K_i\,q_i = 0$, $\omega_i^2 = K_i/M_i$.

Proof (sketch): for modes i, j: $[K]\{X\}_i = \omega_i^2[M]\{X\}_i$. Pre-multiply by $\{X\}_j^T$, and write the same with $i \leftrightarrow j$, subtract using symmetry of [M],[K]:

$$(\omega_i^2 - \omega_j^2)\,\{X\}_j^T[M]\{X\}_i = 0 \rightarrow \{X\}_j^T[M]\{X\}_i = 0 \text{ for } \omega_i \neq \omega_j \quad (8a)$$

Modal (normal-co-ordinate) analysis - steps

1. Find ω_1, ω_2 and mode vectors; form modal matrix $[\Phi] = [\,\{X\}^{(1)}\,\{X\}^{(2)}\,]$.
2. Transform $\{x\} = [\Phi]\{q\}$. Then $[\Phi]^T[M][\Phi] = \text{diag}(M_1, M_2)$, $[\Phi]^T[K][\Phi] = \text{diag}(K_1, K_2)$.
3. Solve two SDOF equations $M_i\ddot{q}_i + K_iq_i = Q_i(t)$; initial $q = [\Phi]^{-1}x(0)$.
4. Transform back $\{x\} = [\Phi]\{q\}$.

Matrix (eigenvalue) form

Eqn (5) is the eigenvalue problem $[K]\{X\} = \omega^2[M]\{X\}$ or $[D]\{X\} = (1/\omega^2)\{X\}$ with dynamic matrix $[D] = [K]^{-1}[M]$ (flexibility × mass) - used for iterative (Stodola) methods in Unit IV.

Special case : equal masses, equal springs ($k_1 = k_2 = k_3 = k$)

Frequency equation: $m^2\omega^4 - 4km\omega^2 + 3k^2 = 0 \rightarrow (m\omega^2 - k)(m\omega^2 - 3k) = 0$

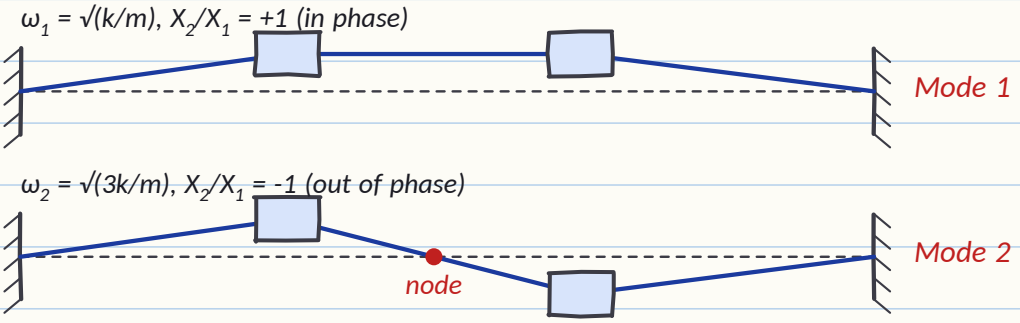


Fig 3.2 Mode shapes (equal masses and springs)

Mode 1: $X_2/X_1 = (2k - m\omega_1^2)/k = 1$; Mode 2: $X_2/X_1 = (2k - 3k)/k = -1$.

Problem 1 (golden-ratio system, free response)

$m_1 = m_2 = 10$ kg, $k_1 = k_2 = 4000$ N/m, $k_3 = 0$ (wall - k_1 - m_1 - k_2 - m_2). Find frequencies, modes; and response if m_1 is displaced 1 mm, m_2 held at 0, released.

Eqn (6) with $k_3 = 0$: $m^2\omega^4 - 3km\omega^2 + k^2 = 0 \rightarrow \omega^2 = (3 \pm \sqrt{5})/2 \cdot k/m = 0.382(400), 2.618(400)$

$\omega_1^2 = 152.8$, $\omega_1 = 12.36$ rad/s (1.97 Hz); $\omega_2^2 = 1047$, $\omega_2 = 32.36$ rad/s (5.15 Hz)

Modes: $r_1 = (2k - m\omega_1^2)/k = 2 - 0.382 = 1.618$; $r_2 = 2 - 2.618 = -0.618$

Initial $x = (1, 0)$: $c_1 + c_2 = 1$; $1.618c_1 - 0.618c_2 = 0 \rightarrow c_1 = 0.2764$, $c_2 = 0.7236$

$x_1 = 0.2764 \cos 12.36t + 0.7236 \cos 32.36t$ (mm)

$x_2 = 0.4472 \cos 12.36t - 0.4472 \cos 32.36t$ (mm)

Symmetric system with coupling spring ($k_1 = k_3 = k$, $k_2 = k_c$, $m_1 = m_2 = m$)

Add and subtract the equations → decoupled in $q_1 = x_1 + x_2$ (in phase) and $q_2 = x_1 - x_2$ (out of phase):

$$\omega_1 = \sqrt{(k/m)} \quad (X_2/X_1 = +1)$$
$$\omega_2 = \sqrt{((k + 2k_c)/m)} \quad (X_2/X_1 = -1) \qquad (8d)$$

Problem 1a (symmetric system)

$m = 5$ kg each; end springs $k = 3000$ N/m; coupling spring $k_c = 2000$ N/m. Find frequencies.

$\omega_1 = \sqrt{(3000/5)} = 24.5$ rad/s (3.90 Hz); $\omega_2 = \sqrt{((3000 + 4000)/5)} = \sqrt{1400} = 37.4$ rad/s (5.96 Hz)

Equations by Lagrange's method (preview of Unit IV)

$T = \tfrac{1}{2}m_1\dot{x}_1^2 + \tfrac{1}{2}m_2\dot{x}_2^2$, $U = \tfrac{1}{2}k_1x_1^2 + \tfrac{1}{2}k_2(x_2 - x_1)^2 + \tfrac{1}{2}k_3x_2^2$. Apply $d/dt(\partial T/\partial\dot{x}_i) - \partial T/\partial x_i + \partial U/\partial x_i = Q_i$:

$i = 1$: $m_1\ddot{x}_1 + k_1x_1 - k_2(x_2 - x_1) = 0$; $i = 2$: $m_2\ddot{x}_2 + k_2(x_2 - x_1) + k_3x_2 = 0$ - identical to eqn (2).

Problem 2 (unequal masses)

$m_1 = 2$ kg, $m_2 = 1$ kg, $k_1 = k_2 = 1000$ N/m, $k_3 = 0$. Find ω_1, ω_2 and modes.

Eqn (6): $2\omega^4 - [2(1000) + 1(2000)]\omega^2 + 10^6 = 0 \rightarrow 2\omega^4 - 4000\omega^2 + 10^6 = 0 \rightarrow \omega^2 =$ $1000(1 - 0.7071)$ and $1000(1 + 0.7071)$

$\omega_1 = 17.11$ rad/s (2.72 Hz), $\omega_2 = 41.32$ rad/s (6.58 Hz)

$r_1 = (2000 - 2\times292.9)/1000 = +1.414$; $r_2 = (2000 - 2\times1707.1)/1000 = -1.414$.

Problem 2a (torsional 2-DOF)

Rotor $J_1 = 2$ kg·m^2 fixed to wall by shaft $k_{t1} = 1000$ N·m/rad; rotor $J_2 = 1$ kg·m^2 connected to J_1 by shaft $k_{t2} = 1000$ N·m/rad. Find natural frequencies.

Mathematically identical to Problem 2 ($J \leftrightarrow m$, $k_t \leftrightarrow k$): $2\omega^4 - 4000\omega^2 + 10^6 = 0 \rightarrow$ $\omega_1 = 17.1$, $\omega_2 = 41.3$ rad/s; $\theta_2/\theta_1 = +1.414$ and -1.414.

Double pendulum (classical 2-DOF example)

Two equal masses m on massless rods of length l hanging one below the other; small angles θ_1, θ_2 (Lagrange or Newton):

$$2\,l\,\ddot{\theta}_1 + l\,\ddot{\theta}_2 + 2\,g\,\theta_1 = 0$$
$$l\,\ddot{\theta}_1 + l\,\ddot{\theta}_2 + g\,\theta_2 = 0 \qquad (8b)$$

With $\lambda = \omega^2 l/g$: $(2 - 2\lambda)\Theta_1 - \lambda\Theta_2 = 0$, $-\lambda\Theta_1 + (1 - \lambda)\Theta_2 = 0 \rightarrow \lambda^2 - 4\lambda + 2 = 0$

$$\omega^2 = (2 - \sqrt{2})\,g/l \text{ and } (2 + \sqrt{2})\,g/l \rightarrow \omega_1 = 0.765\,\sqrt{(g/l)},\ \omega_2 = 1.848\,\sqrt{(g/l)}$$
$$\Theta_2/\Theta_1 = (2 - 2\lambda)/\lambda = +1.414 \text{ (mode 1)}, -1.414 \text{ (mode 2)} \qquad (8c)$$

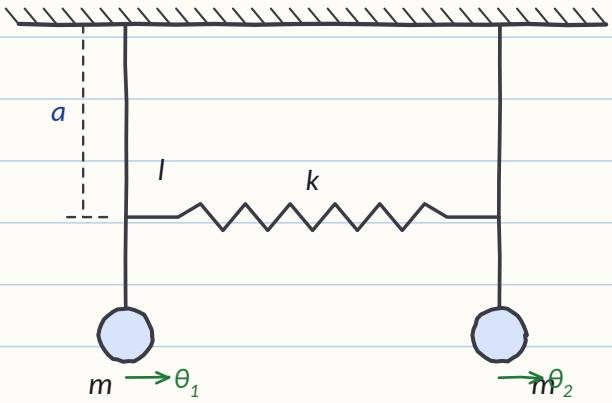
For $l = 1$ m: $\omega_1 = 2.40$ rad/s, $\omega_2 = 5.79$ rad/s. In mode 1 both swing the same way (lower bob moves more); in mode 2 they swing opposite, with a node on the upper rod.

Torsional two-rotor, three-segment systems

Same method with $m \rightarrow J$ (inertia), $k \rightarrow k_t = GJ/l$, $x \rightarrow \theta$: $[J]\{\ddot{\theta}\} + [k_t]\{\theta\} = 0$. If a shaft end is free (no restraint) one root is $\omega = 0$ (rigid-body rotation); semi-definite system. e.g. two rotors: $\omega_n = \sqrt{(k_t(J_1 + J_2)/(J_1J_2))}$ (Unit I).

4. Coupled Pendulum

Two identical simple pendulums (mass m, length l) joined by a spring k attached at distance a below pivot. Small angles θ_1, θ_2.



$$m l^2 \ddot{\theta}_1 + m g l \theta_1 + k a^2(\theta_1 - \theta_2) = 0$$
$$m l^2 \ddot{\theta}_2 + m g l \theta_2 - k a^2(\theta_1 - \theta_2) = 0$$

Fig 3.3 Coupled pendulum

Equations of motion (moments about pivot, small θ):

$$m l^2 \ddot{\theta}_1 + (m g l + k a^2)\theta_1 - k a^2 \theta_2 = 0$$
$$m l^2 \ddot{\theta}_2 - k a^2 \theta_1 + (m g l + k a^2)\theta_2 = 0 \quad (9)$$

Add and subtract (principal co-ordinates $q_1 = \theta_1 + \theta_2$, $q_2 = \theta_1 - \theta_2$):

$$\ddot{q}_1 + (g/l) q_1 = 0 \rightarrow \omega_1 = \sqrt{(g/l)} \text{ (in phase; spring not stretched)}$$
$$\ddot{q}_2 + (g/l + 2ka^2/ml^2) q_2 = 0 \rightarrow \omega_2 = \sqrt{(g/l + 2ka^2/ml^2)} \text{ (out of phase)} \quad (10)$$

- Mode 1: $\theta_1 = \theta_2$ - pendulums swing together at the free-pendulum frequency.
- Mode 2: $\theta_1 = -\theta_2$ - spring stretches/compresses; higher frequency. If a = l (spring at bob): $\omega_2^2 = g/l + 2k/m$.

Beating - energy transfer

Start with $\theta_1(0) = \theta_0$, $\theta_2(0) = 0$, released from rest: $\theta_1 = (\theta_0/2)(\cos \omega_1 t + \cos \omega_2 t)$, $\theta_2 = (\theta_0/2)(\cos \omega_1 t - \cos \omega_2 t)$

$$\theta_1 = \theta_0 \cos[(\omega_2 - \omega_1)t/2] \cos[(\omega_2 + \omega_1)t/2]$$
$$\theta_2 = \theta_0 \sin[(\omega_2 - \omega_1)t/2] \sin[(\omega_2 + \omega_1)t/2] \quad (11)$$

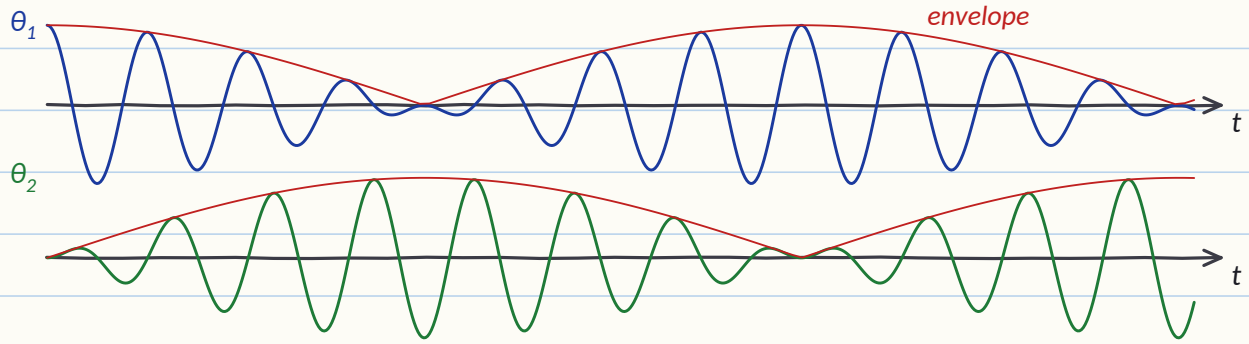


Fig 3.4 Beats: energy passes back and forth between pendulums

- *Pendulum 1 loses amplitude as 2 gains; complete transfer takes $t = \pi/(\omega_2 - \omega_1)$. Weak coupling (small k) → slow beats.*

Problem 3 (coupled pendulum)

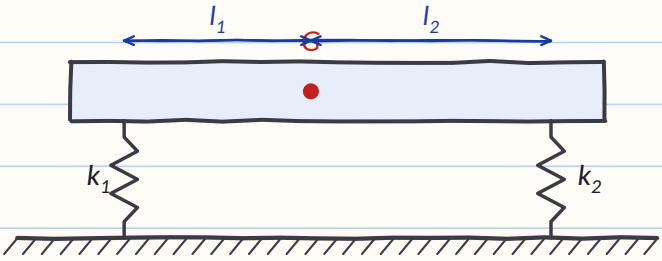
l = 1 m, m = 1 kg, spring k = 2 N/m attached at the bobs ($a = l$). Find ω_1, ω_2 and time for complete energy transfer.

$\omega_1 = \sqrt{(9.81/1)} = 3.132$ rad/s; $\omega_2 = \sqrt{(9.81 + 2\times2/1)} = \sqrt{13.81} = 3.716$ rad/s

$t = \pi/(\omega_2 - \omega_1) = \pi/0.584 = 5.38$ s.

5. Coordinate Coupling - Vehicle Bounce and Pitch

Rigid car body (mass m, inertia J about C.G.) on front and rear suspensions k_1, k_2 at distances l_1, l_2 from C.G. Co-ordinates: x (bounce of C.G.), θ (pitch).



$$m\ddot{x} + (k_1+k_2)x - (k_1l_1 - k_2l_2)\theta = 0$$

$$J\ddot{\theta} - (k_1l_1 - k_2l_2)x + (k_1l_1^2 + k_2l_2^2)\theta = 0$$

coupling term: $k_1l_1 - k_2l_2$

Fig 3.5 Bounce-pitch model

- *If $k_1l_1 = k_2l_2$ the equations are uncoupled (static coupling vanishes): $\omega_x = \sqrt{((k_1+k_2)/m)}$, $\omega_\theta = \sqrt{((k_1l_1^2+k_2l_2^2)/J)}$.*
- *Otherwise bounce and pitch always occur together; the system oscillates about two nodes (centres of oscillation). Poor ride comfort when node lies inside the wheelbase.*
- *Coupling depends on chosen co-ordinates; using co-ordinates measured at a point other than G can give dynamic coupling ($m_{12} \neq 0$) instead.*

Problem 4 (uncoupled case)

$m = 1200$ kg, $J = 1500$ kg·m^2, $l_1 = 1.2$ m, $l_2 = 1.6$ m, $k_1 = 24$ kN/m, $k_2 = 18$ kN/m. Check coupling, find frequencies.

$k_1 l_1 = 28\,800 = k_2 l_2 = 28\,800 \rightarrow$ uncoupled. $\omega_x = \sqrt{(42000/1200)} = 5.92$ rad/s (0.94 Hz)

$\omega_\theta = \sqrt{((24000\times1.44 + 18000\times2.56)/1500)} = \sqrt{53.76} = 7.33$ rad/s (1.17 Hz)

6. Forced Vibration of Two-DOF System (undamped)

Harmonic force $F_1 = F_0 \sin \omega t$ on mass m_1; none on m_2.

$$m_1 \ddot{x}_1 + (k_1 + k_2)\, x_1 - k_2\, x_2 = F_0 \sin \omega t$$
$$m_2 \ddot{x}_2 - k_2\, x_1 + (k_2 + k_3)\, x_2 = 0 \qquad (12)$$

Steady state: $x_i = X_i \sin \omega t$:

$$(k_1+k_2 - m_1\omega^2)\, X_1 - k_2\, X_2 = F_0$$
$$-k_2\, X_1 + (k_2+k_3 - m_2\omega^2)\, X_2 = 0 \qquad (13)$$

Cramer's rule with $\Delta(\omega) = (k_1+k_2 - m_1\omega^2)(k_2+k_3 - m_2\omega^2) - k_2^2$:

$$X_1 = (k_2 + k_3 - m_2\omega^2)\, F_0 / \Delta(\omega)$$
$$X_2 = k_2\, F_0 / \Delta(\omega) \qquad (14)$$

- $\Delta(\omega) = 0$ is the frequency equation $\rightarrow$ two resonances, at $\omega = \omega_1$ and $\omega = \omega_2$.
- If $k_2 + k_3 - m_2\omega^2 = 0 \rightarrow X_1 = 0$ (anti-resonance): m_1 stands still although it carries the force! Basis of the vibration absorber.
- Between the two resonances the masses move out of phase; beyond ω_2 amplitudes fall. Phase changes by 180° across each resonance.

Problem 6 (forced response, numbers)

Use the system of Problem 1 ($m_1 = m_2 = 10$ kg, $k_1 = k_2 = 4000$ N/m, $k_3 = 0$) with $F_0 = 20$ N on m_1. Find X_1, X_2 at $\omega = 10$ and $\omega = 20$ rad/s.

$\Delta(\omega) = (8000 - 10\omega^2)(4000 - 10\omega^2) - 4000^2$

$\omega = 10$: $\Delta = (7000)(3000) - 1.6\times10^7 = 5\times10^6$; $X_1 = (3000)(20)/\Delta = 12$ mm, $X_2 = 4000(20)/\Delta = 16$ mm (in phase, below $\omega_1 = 12.36$).

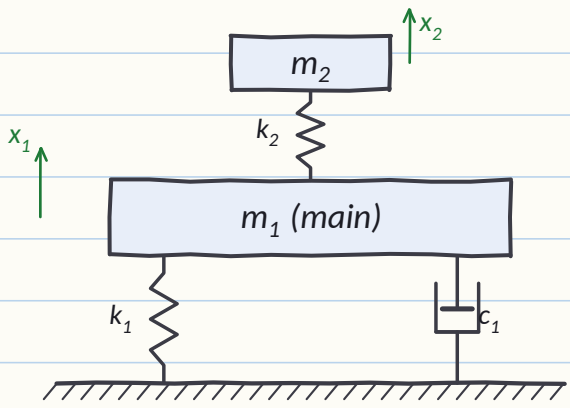
$\omega = 20$: $4000 - 10(400) = 0 \rightarrow X_1 = 0$ (anti-resonance), $X_2 = k_2 F_0/\Delta = 4000(20)/(-1.6\times10^7) = -5$ mm: m_2 acts as an absorber; its spring force $k_2 X_2 = -20$ N $= -F_0$.

Damped forced response (outline)

- Equations: $[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{F\}e^{i\omega t} \rightarrow \{X\} = [K - \omega^2 M + i\omega C]^{-1}\{F\}$ (matrix of mechanical impedance $Z(\omega) = K - \omega^2 M + i\omega C$).
- Damping removes the infinite peaks at ω_1, ω_2 and fills the anti-resonance ($X_1 \neq 0$ exactly). With proportional damping $[C] = \alpha[M] + \beta[K]$ the modal method still decouples.

7. Dynamic Vibration Absorber (DVA)

A machine (m_1, k_1) is excited by force $F_0 \sin \omega t$ at (or near) its resonance → large vibration. Attach an auxiliary spring-mass (m_2, k_2) - the dynamic vibration absorber - to cancel it. Becomes a 2-DOF system.



Absorber tuned: $\omega_a = \sqrt{(k_2/m_2)} = \omega$

Main mass: $X_1 = 0$

Absorber: $X_2 = -F_0/k_2$

Fig 3.6 Dynamic vibration absorber

Equations (main system with $k_3 = 0$, damping neglected) - same as (12) with $k_1 \rightarrow k_1$, k_2, and no k_3:

$$m_1 \ddot{x}_1 + (k_1 + k_2) x_1 - k_2 x_2 = F_0 \sin \omega t$$
$$m_2 \ddot{x}_2 + k_2(x_2 - x_1) = 0 \qquad (15)$$

From (14) with $k_3 = 0$:

$$X_1 = (k_2 - m_2\omega^2) F_0 / [\, (k_1+k_2 - m_1\omega^2)(k_2 - m_2\omega^2) - k_2^2 \,]$$
$$X_2 = k_2 F_0 / [\text{ same denominator }] \qquad (16)$$

Tuned absorber : $k_2 - m_2\omega^2 = 0$, i.e. $\omega_a = \sqrt{(k_2/m_2)} = \omega$ (absorber natural frequency = excitation frequency):

$$X_1 = 0,\ X_2 = -F_0/k_2$$

spring force on main mass: $k_2X_2 = -F_0$ - exactly cancels exciting force. (17)

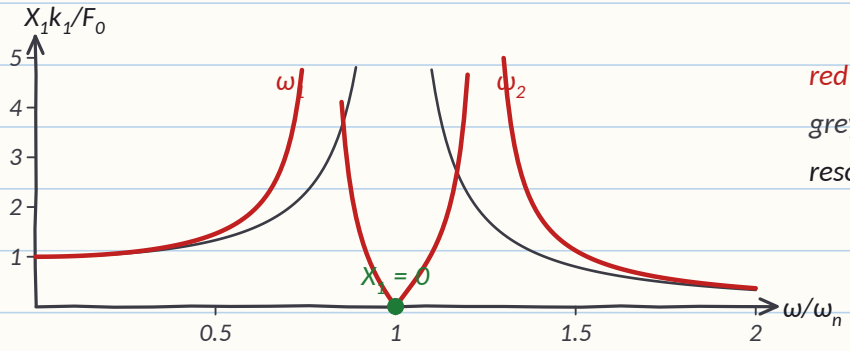
- Design: choose $k_2 = F_0/X_{2,allowed}$ (limits absorber stroke); then $m_2 = k_2/\omega^2$.
- Mass ratio $\mu = m_2/m_1$. Typical μ = 0.05-0.25. Absorber should not be so light that its stroke is excessive.

Natural frequencies after adding absorber ($\omega_a = \omega_n = \sqrt{(k_1/m_1)}$)

Let $s = (\omega/\omega_n)^2$, $k_2/k_1 = \mu$. Denominator of (16) = 0 → $s^2 - (2 + \mu)s + 1 = 0$

$$(\omega/\omega_n)^2 = 1 + \mu/2 - \sqrt{(\mu + \mu^2/4)} \text{ and } 1 + \mu/2 + \sqrt{(\mu + \mu^2/4)} \quad (18)$$

- Original resonance at ω_n is replaced by two resonances, one below and one above ω_n. Larger μ → wider separation → wider operating band.
- Drawback: at speeds other than the tuned speed (e.g. while starting/stopping) the machine passes through one of the new resonances; works for narrow speed range.



red : with absorber (μ = 0.2)

grey: without absorber

resonances at 0.80 ω_n, 1.25 ω_n

Fig 3.7 Main-mass response with tuned absorber

Problem 5 (design of absorber)

Machine m_1 = 200 kg, k_1 = 80 kN/m, force 100 N at ω = 20 rad/s (resonance, $\omega_n = \sqrt{(80000/200)} = 20$). Absorber stroke limited to 5 mm. Find k_2, m_2, new natural frequencies.

$k_2 = F_0/X_2 = 100/0.005 =$ 20 000 N/m; $m_2 = k_2/\omega^2 = 20000/400 =$ 50 kg (μ = 0.25, ω_a = 20 rad/s)

$(\omega/\omega_n)^2 = 1.125 - 0.515 = 0.610$ and $1.125 + 0.515 = 1.640$

$\omega_1 = 20\sqrt{0.610} = 15.6$ rad/s; $\omega_2 = 20\sqrt{1.640} = 25.6$ rad/s - operate outside 15.6-25.6 range except at tuned speed.

Damped absorber (Den Hartog optimum)

- Absorber with viscous damper c_2: amplitude of main mass remains finite over a wide frequency band (no zero, but no infinite peaks).
- Optimum tuning: $f_{opt} = \omega_a/\omega_n = 1/(1 + \mu)$; optimum damping: $\zeta_{opt} = \sqrt{[\ 3\mu / (8(1 + \mu)^3)\ ]}$.
- Examples: pendulum absorbers in skyscrapers (tuned mass dampers), engine crankshaft torsional dampers, power-line stockbridge dampers, bladed disks.

Applications of 2-DOF theory

- Vehicle suspension (quarter-car: sprung + unsprung mass), bounce-pitch ride analysis.
- Tuned mass dampers on tall buildings and bridges; crankshaft torsional dampers; stockbridge dampers on transmission lines.
- Machine tool: spindle + tool holder; rotor-bearing systems; two-storey shear building (m_1, m_2, storey stiffnesses k_1, k_2).
- Vibration absorbers for helicopters, washing machines, ship propeller shafts.

8. Quick Revision & Short Questions

Item	Result
Equation of motion	$[M]\{\ddot{x}\} + [K]\{x\} = 0$; assume $x = X\cos(\omega t + \varphi)$
Frequency equation	$\det[K - \omega^2 M] = 0 \rightarrow$ quadratic in ω^2
Mode shape ratio	$X_2/X_1 = (k_1+k_2 - m_1\omega^2)/k_2$
Equal m, k (3 springs)	$\omega_1 = \sqrt{(k/m)}$ (1 : 1); $\omega_2 = \sqrt{(3k/m)}$ (1 : -1)
Coupled pendulum	$\omega_1 = \sqrt{(g/l)}$; $\omega_2 = \sqrt{(g/l + 2ka^2/ml^2)}$
Anti-resonance	$X_1 = 0$ when $\omega^2 = k_2/m_2$
DVA	$\omega_a = \omega$; $X_2 = -F_0/k_2$; new ω from $s^2 - (2+\mu)s + 1 = 0$

1. What is a principal mode? A vibration in which every point moves harmonically at the same frequency, with fixed amplitude ratio.
2. What is a node? A point of zero displacement in a mode; mode 2 of a 2-DOF system has one node.
3. Static vs dynamic coupling? Off-diagonal terms in [K] vs [M]; depends on co-ordinates chosen.

4. **What are normal co-ordinates?** Co-ordinates in which both [M] and [K] become diagonal; each equation is a SDOF equation.
5. **Why does a DVA work only near one frequency?** It is tuned to $\omega_a = \omega$; at other frequencies absorber force no longer cancels F_0.
6. **How many natural frequencies does a 2-DOF have?** Two; semi-definite systems have one zero (rigid-body) frequency.
7. **Why are modes orthogonal?** Because [M],[K] are symmetric and $\omega_i \neq \omega_j$; $\{X\}_i^T[M]\{X\}_j = 0$ - each mode can be treated independently.
8. **What is a semi-definite system?** One with a rigid-body mode ($\omega = 0$), e.g. two rotors on a free shaft, a train of cars.
9. **Effect of mass ratio on DVA?** Larger μ widens the band between the two new resonances (more robust) but adds more mass.
10. **What happens at anti-resonance?** The main mass has zero steady amplitude; absorber spring force balances the exciting force.

Important questions: (1) derive natural frequencies & modes of 2-DOF spring-mass system (2) coupled pendulum with beats (3) forced response & anti-resonance (4) design a dynamic vibration absorber (5) bounce-pitch coupling of a vehicle.

UNIT - 4

Multi Degree Freedom Systems

Subject: Mechanical Vibrations | Code: 23A03702a

Syllabus: Lagrangian method for formulation of equation of motion, influence co-efficient method, lumped mass and distributed mass systems, Stodola method, Holzer's method, modal analysis of free and forced vibrations.

1. Multi-Degree-of-Freedom (MDOF) Systems

System needing n independent co-ordinates. Equations of motion are n coupled 2nd-order ODEs - written in matrix form:

$$[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{F(t)\} \quad (1)$$

- $[M]$ = mass matrix, $[K]$ = stiffness matrix, $[C]$ = damping matrix, all $n \times n$, symmetric. $\{x\}$, $\{F\}$ are $n \times 1$ vectors.
- Free undamped vibration: assume $\{x\} = \{X\}\cos(\omega t - \varphi) \rightarrow [K]\{X\} = \omega^2[M]\{X\}$ (eigenvalue problem).
- Non-trivial solution: $|[K] - \omega^2[M]| = 0 \rightarrow$ polynomial of degree n in $\omega^2 \rightarrow n$ natural frequencies $\omega_1 \le \omega_2 \le \ldots \le \omega_n$, and n mode shapes $\{X\}^{(i)}$.
- Exact solution for $n > 3$ by hand is impractical → approximate methods (Rayleigh, Stodola, Holzer, Dunkerley) or computer.

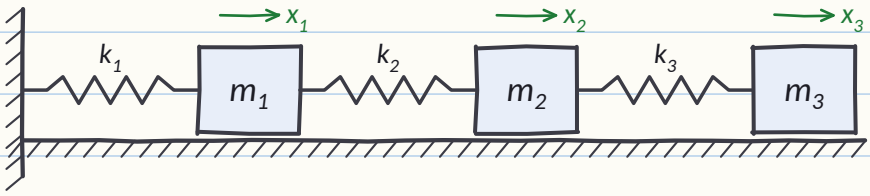


Fig 4.1 Three-DOF chain

2. Lagrangian Method for Equations of Motion

Energy method suited to complex systems (no free-body diagrams, no constraint forces). Uses generalised co-ordinates q_i (independent co-ordinates, number = DOF; any units - length, angle).

Lagrange's equation for $i = 1 \dots n$:

$$d/dt\,(\partial T/\partial \dot{q}_i) - \partial T/\partial q_i + \partial D/\partial \dot{q}_i + \partial U/\partial q_i = Q_i \quad (2)$$

- T = kinetic energy, U = potential energy, D = Rayleigh dissipation function = $\tfrac{1}{2}\sum c_{ij}\dot{q}_i\dot{q}_j$ ($\tfrac{1}{2} c \dot{x}^2$ for dashpot), Q_i = generalised non-conservative force.
- Quadratic forms: $T = \tfrac{1}{2}\{\dot{q}\}^T[M]\{\dot{q}\}$, $U = \tfrac{1}{2}\{q\}^T[K]\{q\}$ → coefficients give $[M]$ and $[K]$ directly: $m_{ij} = \partial^2 T/\partial\dot{q}_i\partial\dot{q}_j$, $k_{ij} = \partial^2 U/\partial q_i \partial q_j$.

Procedure

1. Select generalised co-ordinates $q_1 \dots q_n$.
2. Write T, U (and D, Q_i) in terms of q and $\dot{q}$.
3. Differentiate as per (2) for each q_i → n equations.
4. Arrange in matrix form $[M]\{\ddot{q}\} + [C]\{\dot{q}\} + [K]\{q\} = \{Q\}$.

Example - three-mass chain (Fig 4.1, equal m and k):

$T = \tfrac{1}{2}m(\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2)$, $U = \tfrac{1}{2}k[\,x_1^2 + (x_2 - x_1)^2 + (x_3 - x_2)^2\,]$

$\partial U/\partial x_1 = k(2x_1 - x_2)$; $\partial U/\partial x_2 = k(-x_1 + 2x_2 - x_3)$; $\partial U/\partial x_3 = k(-x_2 + x_3)$

$$m\,[\,1\,0\,0\,;\,0\,1\,0\,;\,0\,0\,1\,]\,\{\ddot{x}\} + k\,[\,2\,{-1}\,0\,;\,{-1}\,2\,{-1}\,;\,0\,{-1}\,1\,]\,\{x\} = \{0\} \quad (3)$$

Example 2 - spring-pendulum (mass m on spring k, which swings): use r (spring length) and θ: $T = \tfrac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$, $U = \tfrac{1}{2}k(r - l_0)^2 - m\,g\,r\cos\theta$. Lagrange gives the two coupled non-linear equations $m\,\ddot{r} - m\,r\dot{\theta}^2 + k(r - l_0) - m\,g\cos\theta = 0$, $m\,r^2\ddot{\theta} + 2\,m\,r\,\dot{r}\,\dot{\theta} + m\,g\,r\sin\theta = 0$.

Advantage:

Lagrange's method: scalar energies only → easier than Newton for MDOF/constrained systems; constraint forces never appear.

Problem 1a (double pendulum by Lagrange)

Masses m_1, m_2; rods l_1, l_2; co-ordinates θ_1, θ_2 (from vertical).

$T = \tfrac{1}{2}(m_1 + m_2)l_1^2\dot{\theta}_1^2 + \tfrac{1}{2}m_2 l_2^2\dot{\theta}_2^2 + m_2 l_1 l_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)$

$U = -(m_1 + m_2)\,g\,l_1\cos\theta_1 - m_2\,g\,l_2\cos\theta_2$

Small angles ($\cos \approx 1$, $\sin \approx \theta$, drop 2nd-order terms) - Lagrange gives:

$$\begin{aligned}(m_1+m_2)\,l_1^2\,\ddot{\theta}_1 + m_2\,l_1 l_2\,\ddot{\theta}_2 + (m_1+m_2)\,g\,l_1\,\theta_1 &= 0\\ m_2\,l_1 l_2\,\ddot{\theta}_1 + m_2\,l_2^2\,\ddot{\theta}_2 + m_2\,g\,l_2\,\theta_2 &= 0\end{aligned} \quad (3a)$$

For $m_1 = m_2$, $l_1 = l_2$ these reduce to the double-pendulum equations of Unit III: $\omega^2 = (2 - \sqrt{2})g/l$ and $(2 + \sqrt{2})g/l$.

3. Influence Coefficient Method

Describes the elastic properties of the system directly from static behaviour. Two forms:

(a) Stiffness influence coefficients k_{ij}

k_{ij} = force required at point i to produce unit displacement at j, with all other points held at zero displacement. Force-displacement: $\{F\} = [K]\{x\}$

$$F_i = \Sigma_j\, k_{ij}\, x_j \quad [K] = [k_{ij}] \text{ (symmetric: } k_{ij} = k_{ji}) \tag{4}$$

Three-mass chain: give $x_1 = 1$, $x_2 = x_3 = 0 \rightarrow$ forces needed: $k_{11} = 2k$, $k_{21} = -k$, $k_{31} = 0$ ... $\rightarrow$ matrix of eqn (3).

(b) Flexibility influence coefficients a_{ij}

a_{ij} = deflection at i due to a unit load at j (no other loads). $\{x\} = [a]\{F\}$

$$x_i = \Sigma_j\, a_{ij}\, F_j \quad [a] = [K]^{-1},\ a_{ij} = a_{ji} \text{ (Maxwell's reciprocal theorem)} \tag{5}$$

For free vibration the force on mass j is the inertia force $-m_j\ddot{x}_j$:

$$\{x\} = -[a][M]\{\ddot{x}\} \rightarrow [a][M]\{X\} = (1/\omega^2)\{X\} \text{ i.e. } [D]\{X\} = \lambda\{X\},\ \lambda = 1/\omega^2$$
$$[D] = [a][M] = [K]^{-1}[M] \text{ (dynamic matrix)} \tag{6}$$

- Flexibility form is preferred when beams/shafts with several masses are given (deflections easy to get from beam formulae); it is the starting point of Stodola's method.

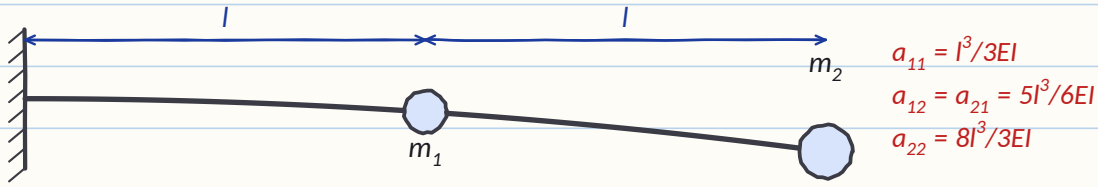


Fig 4.2 Cantilever with two masses - flexibility coefficients

Deflection of cantilever at x due to load P at a ($x \geq a$): $y = P\,a^2(3x - a)/6EI$. With $a = l$: $a_{11} = l^3/3EI$; $x = 2l$: $a_{21} = 5l^3/6EI$; $a = x = 2l$: $a_{22} = 8l^3/3EI$.

Problem 1 (two masses on cantilever)

Massless cantilever (EI constant), $m_1 = m_2 = m$ at l and 2l. Find ω_1, ω_2.

$[a][M] = (m\,l^3/6EI)\ [\ 2\ 5\ ;\ 5\ 16\ ]$. Let $\lambda = 1/\omega^2$, $\mu_0 = m\,l^3/6EI$: $|\ 2\mu_0 - \lambda\ ,\ 5\mu_0\ ;\ 5\mu_0\ ,\ 16\mu_0 - \lambda\ | = 0$

$\lambda^2 - 18\mu_0\lambda + 7\mu_0^2 = 0 \rightarrow \lambda = \mu_0(9 \pm \sqrt{74}) = 17.60\mu_0,\ 0.3977\mu_0$

$$\omega_1^2 = 1/17.60\mu_0 = 0.3409\ EI / (m\ l^3) \rightarrow \omega_1 = 0.584\ \sqrt{(EI/ml^3)}$$
$$\omega_2^2 = 1/0.3977\mu_0 = 15.09\ EI / (m\ l^3) \rightarrow \omega_2 = 3.885\ \sqrt{(EI/ml^3)} \quad (7)$$

Check: $6/17.60 = 0.3409$; $6/0.3977 = 15.09$. Mode 1 ratio $X_2/X_1 = (\lambda - 2\mu_0)/5\mu_0 = 3.12$.

Problem 1b (simply supported beam, two masses at third points)

Massless beam, span l, equal masses m at l/3 and 2l/3. Flexibility coefficients: $a_{11} = a_{22} = 4l^3/243EI = 8/486\ (l^3/EI)$, $a_{12} = a_{21} = 7l^3/486EI$.

$[D] = m(l^3/486EI)[\ 8\ 7\ ;\ 7\ 8\ ]$; $\lambda = 1/\omega^2 = (m\ l^3/486EI)(8 \pm 7) = 15,\ 1\ (\times\ m\ l^3/486EI)$

$$\omega_1 = \sqrt{(486/15)}\ \sqrt{(EI/ml^3)} = 5.69\ \sqrt{(EI/ml^3)}\ \text{mode}\ (1,\ 1)$$
$$\omega_2 = \sqrt{486}\ \sqrt{(EI/ml^3)} = 22.05\ \sqrt{(EI/ml^3)}\ \text{mode}\ (1,\ -1) \quad (7a)$$

Bounds on fundamental frequency (3-mass chain, equal m, k)

Rayleigh (trial shape X = (1, 2, 3)): $\{X\}^T[K]\{X\} = 3k$, $\{X\}^T[M]\{X\} = 14m \rightarrow \omega_1^2 \approx 0.2143$ k/m (upper bound).

Dunkerley: $1/\omega_1^2 \approx \Sigma\ a_{ii}m_i = (1 + 2 + 3)\ m/k = 6m/k \rightarrow \omega_1^2 \approx 0.1667$ k/m (lower bound).

Exact: 0.1981 k/m, so $0.1667 < 0.1981 < 0.2143$ - the two simple methods bracket the true value.

4. Lumped Mass and Distributed Mass Systems

- Lumped (discrete) mass system: mass concentrated at finite points joined by massless elastic elements. Finite DOF, matrix equations (ODEs).
- Distributed (continuous) system: mass & elasticity spread continuously (beams, shafts, strings, rods). Infinite DOF; governed by partial differential equations; infinite natural frequencies.
- Lumping: a continuous system is approximated by lumped masses at nodes (finite-element idea). Accuracy improves with more masses; lowest few frequencies are good, higher ones poor.

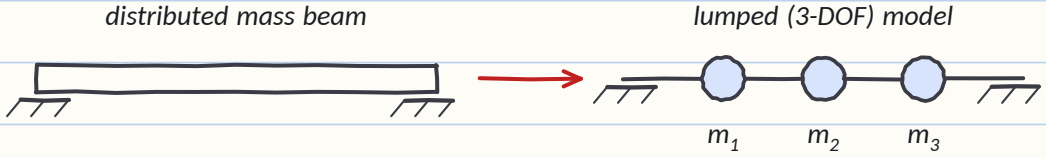


Fig 4.3 Lumping

Continuous member	Natural frequencies ω_n (rad/s)
String (tension P, mass/length ρA)	$(n\pi/l)\sqrt{(P/\rho A)}$ n = 1, 2, 3...
Bar, longitudinal; fixed-free	$(2n - 1)(\pi/2l)\sqrt{(E/\rho)}$
Bar, longitudinal; fixed-fixed or free-free	$(n\pi/l)\sqrt{(E/\rho)}$
Shaft, torsional; fixed-free	$(2n - 1)(\pi/2l)\sqrt{(G/\rho)}$
Beam, simply supported	$(n\pi/l)^2\sqrt{(EI/\rho A)}$
Beam, cantilever	$(\beta_n l)^2/l^2\sqrt{(EI/\rho A)}$; $\beta_1 l = 1.875$, $\beta_2 l = 4.694$, $\beta_3 l =$ 7.855
Beam, fixed-fixed	same form, $\beta_1 l = 4.730$, $\beta_2 l = 7.853$, $\beta_3 l = 10.996$

Rayleigh's quotient (fundamental frequency estimate)

Assume a mode shape y_i (static deflection curve is excellent). From $T_{max} = U_{max}$:

$$\omega_1^2 \approx g \Sigma W_i y_i / \Sigma W_i y_i^2 = \Sigma m_i g y_i / \Sigma m_i y_i^2 \; (= \{X\}^T[K]\{X\} / \{X\}^T[M]\{X\}) \quad (8)$$

- Always gives ω_1 higher than (or equal to) the true value (upper bound); Dunkerley (Unit I) gives lower bound - the true value lies between.

5. Stodola Method (Matrix Iteration)

Iterative method that converges to the fundamental mode (lowest ω) first, using the dynamic matrix $[D] = [a][M]$: $[D]\{X\} = (1/\omega^2)\{X\}$.

Procedure (fundamental mode)

1. Form $[D] = [a][M]$ (flexibility matrix × mass matrix).
2. Assume a trial mode $\{X\}_1$ (e.g. all 1's, or the expected shape).
3. Compute $\{X\}_2' = [D]\{X\}_1$. Normalise (divide by its first or largest element) to get $\{X\}_2$.
4. Repeat: $\{X\}_{n+1}' = [D]\{X\}_n$. The normalising factor converges to $\lambda_1 = 1/\omega_1^2$, and $\{X\}_n$ to the first mode shape.
5. Stop when successive λ and mode vectors agree to required accuracy. $\omega_1 = 1/\sqrt{\lambda_1}$.

- Higher modes: remove the lower mode by sweeping (orthogonality) $[D]_2 = [D]([I] - \{X\}_1\{X\}_1^T[M] / M_1)$, $M_1 = \{X\}_1^T[M]\{X\}_1$, and iterate again → ω_2, etc. (Round-off error spoils accuracy as higher modes are reached.)
- Why it works: any trial vector = $\Sigma c_i\{X\}^{(i)}$; each multiplication by $[D]$ multiplies

mode i by $\lambda_i = 1/\omega_i^2$; the largest λ (lowest ω) soon dominates.

Problem 2 (Stodola on the 3-mass chain)

Equal m, k (Fig 4.1, fixed-free). Flexibility matrix of the chain: $a_{ij} = \min(i,j)/k$. So $[D] = (m/k)[\ 1\ 1\ 1\ ;\ 1\ 2\ 2\ ;\ 1\ 2\ 3\ ]$. Let $\rho = m/k$. Find ω_1.

Iteration	Trial {X}	[D]{X}/ρ	Normalised	Factor
1	1, 1, 1	3, 5, 6	1, 1.667, 2.000	3.000
2	1, 1.667, 2.000	4.667, 8.333, 10.333	1, 1.786, 2.214	4.667
3	1, 1.786, 2.214	5.000, 9.000, 11.214	1, 1.800, 2.243	5.000
4	1, 1.800, 2.243	5.043, 9.086, 11.329	1, 1.802, 2.246	5.043
exact			1, 1.802, 2.247	5.049

$\lambda_1 = 5.049\ m/k \rightarrow \omega_1^2 = 0.1981\ k/m$, $\omega_1 = 0.445\ \sqrt{(k/m)}$ (matches exact $2(1 - \cos \pi/7)$).

For $m = 10\ kg$, $k = 4000\ N/m$: $\omega_1 = 0.445 \times 20 = 8.90\ rad/s$.

Exact frequencies of this chain: $\omega^2 = (k/m)[0.198, 1.555, 3.247] \rightarrow \omega = (0.445, 1.247, 1.802)\sqrt{(k/m)} = 8.90, 24.94, 36.04\ rad/s$; modes (1, 1.802, 2.247), (1, 0.445, -0.802), (1, -1.247, 0.555).

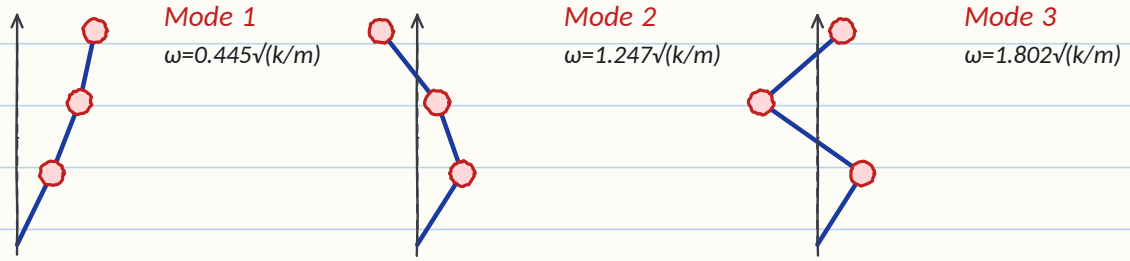
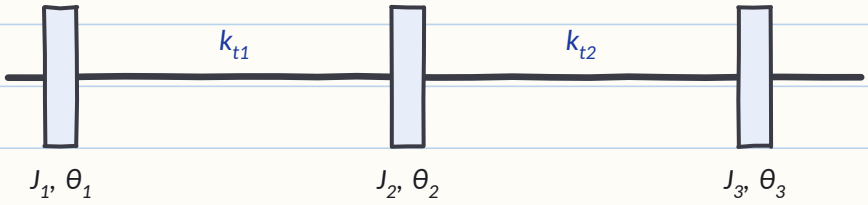


Fig 4.4 Mode shapes of the 3-DOF chain (amplitudes sideways)

6. Holzer's Method

Tabular trial-and-error method for natural frequencies and mode shapes of torsional (also spring-mass) systems with many rotors, in chain form (also geared systems after reduction). Gives any mode, not just the lowest.



free-free: $T_n = 0$ at end

(fixed end: $\theta_n = 0$)

Fig 4.5 Torsional chain for Holzer's method

Principle

At a natural frequency the free system satisfies both end conditions. Free end: zero torque. Assume ω^2, start from one end with $\theta_1 = 1$ and propagate:

$$\text{Inertia torque of rotor } i: T_i = \omega^2 J_i \theta_i$$
$$\text{Shaft torque beyond rotor } i: S_i = \Sigma_{j \le i}\, \omega^2 J_j \theta_j$$
$$\text{Twist of shaft } i: \theta_{i+1} = \theta_i - S_i / k_{t,i} \quad (9)$$

At the last rotor compute residual torque $R(\omega^2) = \Sigma_{all}\, \omega^2 J_j \theta_j$ (free end must be 0). Plot R against ω^2; zeros of R are the natural frequencies (rigid-body mode $\omega = 0$ for free-free).

Steps

1. Choose trial ω^2; $\theta_1 = 1$.
2. Fill table columns: J, θ, $\omega^2 J\theta$, $\Sigma\omega^2 J\theta$ (torque), torque/k_t, next θ.
3. Residual R at end. Repeat for several ω^2; sign change → root nearby; interpolate / refine until $R \approx 0$.
4. The θ values at the root = mode shape. For a fixed end the check is $\theta_n = 0$ instead of $R = 0$.

Problem 3 (three equal rotors, free-free)

$J_1 = J_2 = J_3 = 1$ kg·m^2, $k_{t1} = k_{t2} = 1$ N·m/rad. Find natural frequencies by Holzer.

ω^2	θ_1	Σ after 1	θ_2	Σ after 2	θ_3	Residual
0.5	1	0.500	0.500	0.750	-0.250	0.625
1.0	1	1.000	0.000	1.000	-1.000	0.000
2.0	1	2.000	-1.000	0.000	-1.000	-2.000
3.0	1	3.000	-2.000	-3.000	1.000	0.000

Residual changes sign between 0.5 and 2 and is zero at $\omega^2 = 1$ and 3. Hence $\omega = 0$ (rigid), $\omega_1^2 = 1 \rightarrow \omega = 1$ rad/s, $\omega_2^2 = 3 \rightarrow \omega = 1.732$ rad/s.

Mode shapes (θ_1, θ_2, θ_3): $\omega^2 = 1 \rightarrow (1, 0, -1)$ (node at middle rotor); $\omega^2 = 3 \rightarrow (1, -2,$

1) (two nodes).

- Matches exact eigenvalues 0, 1, 3 of [1 -1 0 ; -1 2 -1 ; 0 -1 1]. For real shafts, k_t = GJ/l and J_i = rotor inertias; geared branches are reduced to one shaft first.

Problem 3a (Holzer, fixed-free spring-mass chain)

Chain of Fig 4.1, m = 1, k = 1 (wall - k - m_1 - k - m_2 - k - m_3). Start at the free end: $x_3 = 1$; $F = \Sigma\, \omega^2 m\, x$; $x_{i-1} = x_i - F/k$. Wall condition: $x_0 = 0$.

ω^2	x_3	x_2	x_1	F at wall	x_0 (residual)
0.1	1	0.900	0.710	0.261	+0.449
0.1981	1	0.802	0.445	0.445	≈ 0
0.3	1	0.700	0.190	0.567	-0.377

Sign change of x_0 between 0.1 and 0.3 → root near 0.198: $\omega_1^2 = 0.1981\ k/m$; mode $(x_1, x_2, x_3) = (0.445, 0.802, 1) \sim (1, 1.802, 2.247)$. Repeating with trial ω^2 near 1.5 and 3.2 gives the higher modes.

7. Modal Analysis - Free and Forced Vibration

Uses orthogonality of modes to convert n coupled equations into n independent SDOF equations. Mode vectors $\{\varphi\}_i$:

$$\{\varphi\}_i^T[M]\{\varphi\}_j = 0,\ \{\varphi\}_i^T[K]\{\varphi\}_j = 0\ (i \neq j)$$
$$\{\varphi\}_i^T[M]\{\varphi\}_i = M_i\ (\text{modal mass}),\ \{\varphi\}_i^T[K]\{\varphi\}_i = K_i = \omega_i^2 M_i \qquad (10)$$

Mass-normalised modes ($M_i = 1$): $[\Phi]^T[M][\Phi] = [I]$, $[\Phi]^T[K][\Phi] = \text{diag}(\omega_i^2)$.

Modal transformation

Put $\{x\} = [\Phi]\{q\}$ (q = principal / modal co-ordinates), pre-multiply by $[\Phi]^T$:

$$M_i \ddot{q}_i + C_i \dot{q}_i + K_i q_i = Q_i(t) = \{\varphi\}_i^T\{F(t)\}\ i = 1\ldots n$$
$$\text{or } \ddot{q}_i + 2\zeta_i\omega_i \dot{q}_i + \omega_i^2 q_i = Q_i/M_i \qquad (11)$$

- Damping decouples only if it is proportional (Rayleigh): $[C] = \alpha[M] + \beta[K] \rightarrow \zeta_i = \tfrac{1}{2}(\alpha/\omega_i + \beta\omega_i)$.
- Each $q_i(t)$ is a SDOF solution (Units I, II); final response $\{x\} = \Sigma\, \{\varphi\}_i\, q_i(t)$ - often only the first few modes are needed.

Free vibration (initial conditions)

Initial modal values: $q_i(0) = \{\varphi\}_i^T[M]\{x_0\} / M_i$, $\dot{q}_i(0) = \{\varphi\}_i^T[M]\{\dot{x}_0\} / M_i$.

$$q_i(t) = q_i(0) \cos \omega_i t + (\dot{q}_i(0)/\omega_i) \sin \omega_i t \quad \{x\} = \Sigma \{\varphi\}_i \, q_i(t) \tag{12}$$

Forced (harmonic) vibration

For $F = \{F_0\} \sin \omega t$: $Q_i = \{\varphi\}_i^T\{F_0\} \sin \omega t$, and (undamped):

$$q_i = (Q_{i0}/M_i) / (\omega_i^2 - \omega^2) \sin \omega t \quad \{x\} = \Sigma \{\varphi\}_i \, q_i \tag{13}$$

- Mode i is strongly excited near $\omega = \omega_i$; a mode is not excited if $\{\varphi\}_i^T\{F_0\} = 0$ (force applied at a node, or orthogonal to the mode).

Problem 4 (modal analysis of a 2-DOF system)

$m_1 = m_2 = 10$ kg, $k_1 = k_2 = k_3 = 4000$ N/m. (a) Free response if $x_0 = (1, 0)$ mm, at rest. (b) Forced response for $F = 20 \sin 10t$ N on m_1.

Modes: $\{\varphi\}_1 = (1, 1)$, $\omega_1^2 = k/m = 400$; $\{\varphi\}_2 = (1, -1)$, $\omega_2^2 = 3k/m = 1200$. $M_1 = M_2 = 20$ kg.

(a) $q_1(0) = \{\varphi\}_1^T M \, x_0/M_1 = 10(1)/20 = 0.5$; $q_2(0) = 0.5 \rightarrow$

$x_1 = 0.5 \cos 20t + 0.5 \cos 34.64t$, $x_2 = 0.5 \cos 20t - 0.5 \cos 34.64t$ (mm) (beats between modes).

(b) $Q_{10} = (1, 1)\cdot(20, 0) = 20$; $Q_{20} = 20$. $q_1 = (20/20)/(400 - 100) = 3.333\times10^{-3}$ m; $q_2 = 1/(1200 - 100) = 0.909\times10^{-3}$ m

$X_1 = q_1 + q_2 = 4.24$ mm, $X_2 = q_1 - q_2 = 2.42$ mm (agrees with direct solution $X_1 = 7000(20)/33\times10^6$).

Problem 5 (orthogonality check, 3-mass chain)

Modes: $\{\varphi\}_1 = (1, 1.802, 2.247)$, $\{\varphi\}_2 = (1, 0.445, -0.802)$, $m = 1$.

$\{\varphi\}_1^T[M]\{\varphi\}_2 = 1 + 1.802(0.445) + 2.247(-0.802) = 1 + 0.802 - 1.802 = 0$ (orthogonal).

Modal masses: $M_1 = 1 + 3.247 + 5.049 = 9.296$ m; $M_2 = 1 + 0.198 + 0.643 = 1.841$ m. Mass-normalised: $\{\varphi\}_1/\sqrt{9.296} = (0.328, 0.591, 0.737)$.

Truncation of modes

- Response is dominated by modes whose frequencies are near the excitation and whose modal force $\{\varphi\}_i^T\{F\}$ is large; high modes respond quasi-statically.
- Mode-acceleration / truncation: keep first few modes (say $p < n$) → big saving for large FE models.

Comparison of approximate methods

Method	Gives	Remark
Rayleigh	ω_1	upper bound; needs a good trial shape
Dunkerley	ω_1	lower bound; needs only a_{ii}
Stodola	ω_1, then higher	iterative; sweeping for higher modes
Holzer	any ω, modes	tabular; chain / torsional systems
Modal analysis	response	exact superposition after eigen-solution

Problem 6 (Lagrange with damping and force)

Masses m_1, m_2; springs k_1 (wall - m_1), k_2 (m_1 - m_2); dashpot c between m_2 and ground; force $F(t)$ on m_2.

$T = \tfrac{1}{2}m_1\dot{x}_1^2 + \tfrac{1}{2}m_2\dot{x}_2^2$; $U = \tfrac{1}{2}k_1x_1^2 + \tfrac{1}{2}k_2(x_2 - x_1)^2$; $D = \tfrac{1}{2}c\,\dot{x}_2^2$; $Q_1 = 0$, $Q_2 = F(t)$.

$$m_1\ddot{x}_1 + (k_1 + k_2)x_1 - k_2x_2 = 0$$
$$m_2\ddot{x}_2 + c\,\dot{x}_2 - k_2x_1 + k_2x_2 = F(t) \qquad (13a)$$

Problem 7 (Rayleigh proportional damping)

For the chain with $\omega_1 = 8.90$, $\omega_2 = 24.94$ rad/s, require $\zeta_1 = 0.02$, $\zeta_2 = 0.03$ in $[C] = \alpha[M] + \beta[K]$, $\zeta_i = \tfrac{1}{2}(\alpha/\omega_i + \beta\omega_i)$.

$0.04 = \alpha/8.90 + 8.90\beta$; $0.06 = \alpha/24.94 + 24.94\beta \rightarrow \beta = 2.10\times10^{-3}$ s, $\alpha = 0.1896$ s^{-1}

Then ζ_3 ($\omega_3 = 36.04$) = $\tfrac{1}{2}(0.1896/36.04 + 0.0021\times36.04) = 0.041$ - higher modes are damped more.

Problem 8 (modal forced response, 3-mass chain)

$m = 10$ kg, $k = 4000$ N/m each; $F = 10 \sin 15t$ N on m_3. Find amplitudes.

$\omega_i^2 = 400 \times (0.1981, 1.555, 3.247) = 79.2, 622.0, 1298.8$; modes (1, 1.802, 2.247), (1, 0.445, -0.802), (1, -1.247, 0.555)

Modal masses $M_i = 10\{\varphi\}^T\{\varphi\} = 92.96, 18.41, 28.63$ kg; modal forces $Q_{i0} = \varphi_i(3)F_0 = 22.47, -8.02, 5.55$ N

$q_i = (Q_{i0}/M_i)/(\omega_i^2 - 225) = -1.658, -1.097, +0.181$ (mm)

$X_1 = \Sigma\varphi_i(1)q_i = -2.58$ mm, $X_2 = -3.70$ mm, $X_3 = -2.75$ mm (negative sign: out of phase with force, since 15 rad/s lies above $\omega_1 = 8.9$).

Check row 1 of equations: $(8000 - 10\times225)(-2.575) - 4000(-3.701) \approx 0$.

8. Quick Revision & Short Questions

Method	Key idea / formula
Lagrange	$d/dt(\partial T/\partial \dot{q}) - \partial T/\partial q + \partial D/\partial \dot{q} + \partial U/\partial q = Q$
Influence coefficients	$\{F\} = [K]\{x\}$, $\{x\} = [a]\{F\}$, $[a] = [K]^{-1}$; $[D] = [a][M]$
Rayleigh quotient	$\omega^2 \approx \{X\}^T[K]\{X\} / \{X\}^T[M]\{X\}$ (upper bound)
Stodola	$\{X\}_{n+1} = [D]\{X\}_n$; factor $\to 1/\omega_1^2$; sweep for higher modes
Holzer	$\theta_{i+1} = \theta_i - \Sigma\omega^2 J\theta/k_t$; residual = 0 at natural frequency
Modal analysis	$\{x\} = [\Phi]\{q\}$; $M_i\ddot{q}_i + K_i q_i = \{\varphi\}_i^T\{F\}$

1. **Why use generalised co-ordinates?** They are independent, automatically satisfy constraints and make Lagrange's equation simple.
2. **State Maxwell's reciprocal theorem.** $a_{ij} = a_{ji}$: deflection at i due to unit load at j equals that at j due to unit load at i.
3. **Which mode does Stodola give first and why?** The fundamental, since [D] has the largest eigenvalue $1/\omega_1^2$.
4. **When does the Holzer residual vanish?** Only at natural frequencies (boundary condition at last rotor satisfied).
5. **Why is modal analysis efficient?** n coupled equations become n SDOF equations; few modes suffice for response.
6. **Lumped vs distributed system?** Finite vs infinite DOF; ODEs vs PDEs.
7. **What is Rayleigh damping?** $[C] = \alpha[M] + \beta[K]$; diagonalised by the undamped modes, so modal equations stay uncoupled.
8. **What is sweeping in Stodola's method?** Using orthogonality to remove an already-found mode from [D] so iteration converges to the next mode.
9. **Rayleigh vs Dunkerley?** Rayleigh gives an upper bound, Dunkerley a lower bound for ω_1.
10. **Number of natural frequencies of a continuous beam?** Infinite (governed by a PDE); lumping retains only n of them.

Applications

- Turbine/generator shaft trains (Holzer), multi-storey buildings (shear-building models), aircraft wing and fuselage (modal testing), machine-tool structures, automobile body-in-white (FE modal analysis).

Formula sheet - Unit IV

- Eigenproblem: $[K]\{X\} = \omega^2[M]\{X\}$; $|[K] - \omega^2[M]| = 0$; $[D] = [K]^{-1}[M] = [a][M]$, $\lambda = 1/\omega^2$.
- Lagrange: $d/dt(\partial T/\partial \dot{q}_i) - \partial T/\partial q_i + \partial D/\partial \dot{q}_i + \partial U/\partial q_i = Q_i$.
- Chain, n equal masses, fixed-free: $\omega_i^2 = 2(k/m)[1 - \cos((2i - 1)\pi/(2n + 1))]$.
- Rayleigh: $\omega^2 \approx \{X\}^T[K]\{X\}/\{X\}^T[M]\{X\}$; Dunkerley: $1/\omega_1^2 \approx \Sigma\, a_{ii}m_i$.
- Holzer: $\theta_{i+1} = \theta_i - (\Sigma_{j \le i}\, \omega^2 J_j\theta_j)/k_{t,i}$; residual = 0 at natural frequency.
- Modal: $\{x\} = [\Phi]\{q\}$; $M_i\ddot{q}_i + C_i\dot{q}_i + K_iq_i = \{\varphi\}_i^T\{F\}$; $q_i(0) = \{\varphi\}_i^T[M]\{x_0\}/M_i$.

Exam tips

- For 2-DOF and 3-DOF: always show frequency equation, roots, mode ratios and a neat sketch of mode shapes with nodes.
- In Stodola/Holzer problems present the iteration/table clearly and state convergence criterion and final ω with units.
- Quote Maxwell's theorem whenever [a] or [K] is shown symmetric; use consistent units (N, m, kg, rad/s).

Important questions: (1) Lagrange's equation applied to a 2/3-DOF system (2) influence coefficients and Maxwell's theorem (3) Stodola iteration problem (4) Holzer table (5) modal analysis of free and forced vibration.

UNIT - 5

Vibration Measurement and Applications

Subject: Mechanical Vibrations | Code: 23A03702a

Syllabus: Transducers: variable resistance, piezoelectric, electrodynamic and LVDT transducers; vibration pickups: vibrometer, accelerometer, velometer and phase distortion; frequency-measuring instruments; vibration exciters: mechanical exciters and electrodynamic shaker.

1. Vibration Measurement - Introduction

Vibration measurement is needed for: (i) finding natural frequencies and damping of structures, (ii) checking operating levels against standards, (iii) condition monitoring and fault diagnosis of machines, (iv) verifying analytical models, (v) field balancing, (vi) testing products (qualification, fatigue).

Quantities measured: displacement, velocity, acceleration, frequency, phase. For harmonic motion $x = X \sin \omega t$: $v = \omega X$, $a = \omega^2 X$ (amplitudes).

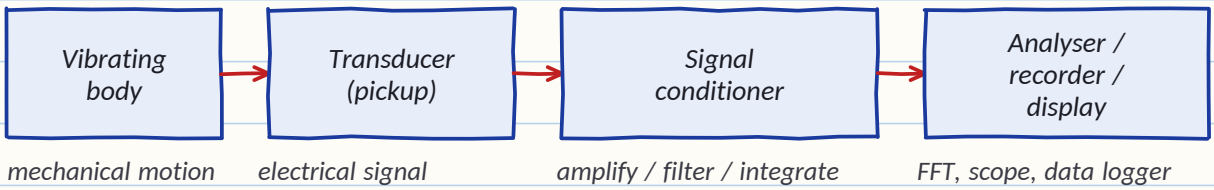


Fig 5.1 Vibration measuring chain

Transducer = device that converts mechanical motion (or force) into an electrical signal (voltage / charge / resistance change).

Classification of transducers

Type	Principle	Examples
Active (self-generating)	Generates its own voltage / charge, needs no external excitation	Piezoelectric, electrodynamic (velocity coil), thermocouple
Passive	Needs external power; motion changes a circuit parameter R, L or C	Strain gauge (R), LVDT (L), capacitive (C), potentiometer (R)

2. Transducers

2.1 Variable resistance transducers (strain gauge)

Electrical resistance of a wire $R = \rho l/A$ changes when it is strained (length and area change, resistivity changes). Gauge factor:

$$GF = (\Delta R/R) / \varepsilon = (\Delta R/R)/(\Delta L/L) \quad (1)$$

metal foil gauge: GF ≈ 2; semiconductor: 50-200

- Bonded foil gauge: fine grid of constantan/nichrome foil on a plastic backing, cemented to the member; measures surface strain (hence stress, force, torque, pressure, acceleration when mounted on a cantilever).
- Resistance change is tiny → measured with a Wheatstone bridge. For quarter bridge (one active gauge) output $e_0 = (E/4)\, GF\, \varepsilon$.
- Also potentiometer (sliding contact on resistor) for low-frequency displacement; unbonded wire gauge accelerometers.
- Advantages: cheap, small, accurate, static and dynamic response (to ~ 10 kHz). Limits: temperature effects (use dummy gauge), low output, fatigue, fragile.

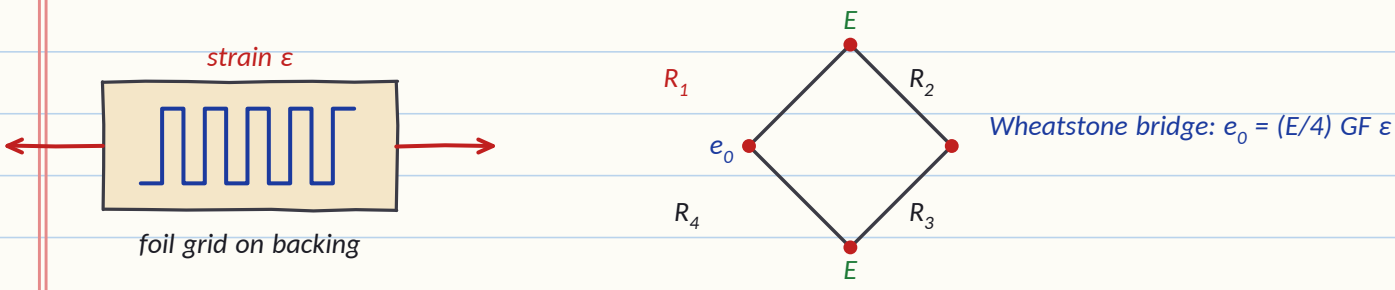


Fig 5.2 Foil strain gauge and bridge

Problem 1 (strain gauge)

A 120 Ω gauge, GF = 2.0, strain 500 με (500×10^{-6}). Find ΔR and quarter-bridge output for E = 10 V.

$\Delta R = R \cdot GF \cdot \varepsilon = 120 \times 2 \times 500\times10^{-6} = 0.12\ \Omega$; $e_0 = (E/4)GF\,\varepsilon = (10/4)(2)(500\times10^{-6}) = 2.5$ mV.

2.2 Piezoelectric transducers

Certain crystals (quartz, Rochelle salt, barium titanate, PZT ceramics) develop an electric charge when mechanically stressed (piezoelectric effect).

$q = d\,F$ (d = charge sensitivity, pC/N; quartz ≈ 2.3 pC/N)

$e_0 = q / C$ (open-circuit voltage; C = crystal + cable + amplifier capacitance)

- Piezo accelerometer: crystal(s) preloaded with a seismic mass; the base acceleration gives force $F = m\,a$ on the crystal → charge ~ acceleration (Fig 5.3).
- Needs a charge amplifier (or ICP built-in amplifier) because of very high output impedance; output affected by cable length if voltage amplifier is used.
- Advantages: very wide frequency range (≈ 1 Hz - 30 kHz), small size, light, rugged, high natural frequency (> 30 kHz), no power (self-generating), good linearity.
- Limits: cannot measure static (DC) acceleration (charge leaks away); temperature and cable noise (triboelectric); high cost.

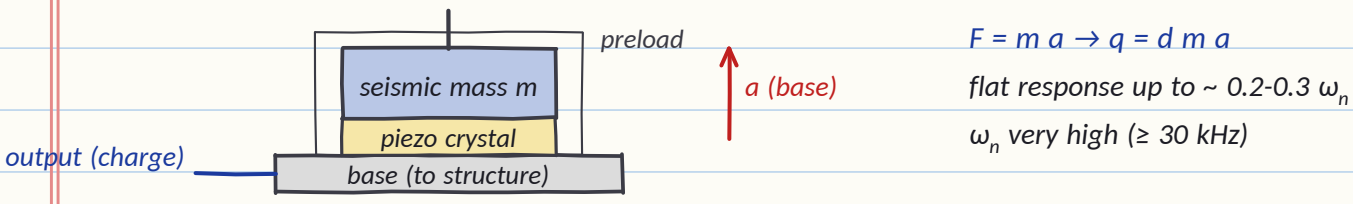


Fig 5.3 Piezoelectric accelerometer (compression type)

Problem 2 (piezo voltage)

Quartz crystal d = 2.3 pC/N loaded by a force amplitude 10 N; total capacitance 100 pF. Find charge and voltage.

$q = dF = 2.3 \times 10 =$ 23 pC; $e_0 = q/C = 23\times10^{-12}/100\times10^{-12} =$ 0.23 V.

2.3 Electrodynamic (electromagnetic) transducers

A coil moves in a magnetic field of a permanent magnet (or a magnet moves inside a coil). By Faraday's law the induced emf is proportional to the velocity:

$$e = B\,l\,v \quad (3)$$

B (T) flux density, l (m) conductor length in field, v (m/s) relative velocity

- Self-generating (active) - no power supply; used in velometers (velocity pickups); output can be integrated to get displacement or differentiated to get acceleration.
- Advantages: robust, low output impedance (little cable problem), high output, no amplifier needed. Limits: bulky, heavy (loads light structures), limited frequency

range (≈ 10 Hz - 1 kHz), sensitive to stray magnetic fields.

- Reverse action - current in the coil in a field produces force F = B l i → basis of the electrodynamic shaker (Section 5).

Example: B = 0.8 T, l = 5 m of wire, v = 0.1 m/s → e = 0.8 × 5 × 0.1 = 0.4 V.

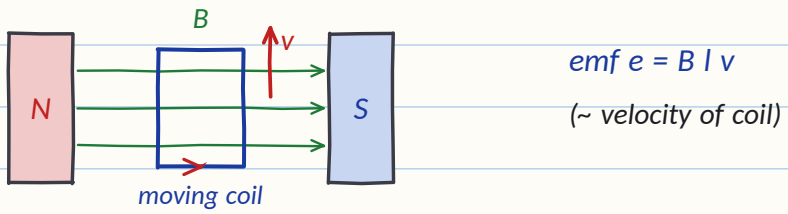


Fig 5.4 Electrodynamic (velocity) transducer

2.4 Linear Variable Differential Transformer (LVDT)

A passive inductive displacement transducer: one primary winding (AC carrier, typically 1-10 kHz) and two identical secondary windings connected in series opposition; a movable ferromagnetic core attached to the vibrating body.

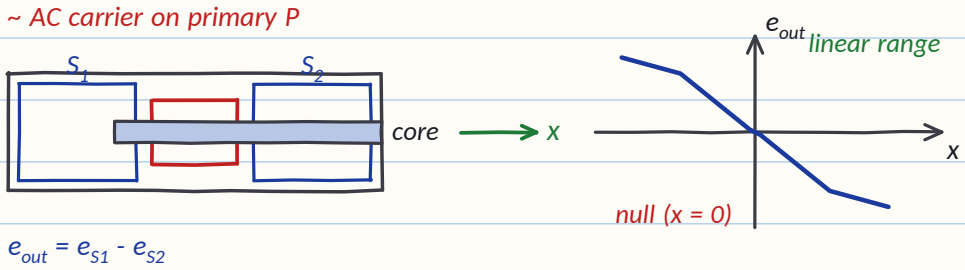


Fig 5.5 LVDT and its output

- Core central (x = 0): equal emf induced in both secondaries → net output e_{out} = 0 (null).
- Core moves by x: one secondary gets more flux → e_{out} ~ x; phase (relative to carrier) tells direction. A phase-sensitive demodulator + low-pass filter gives a DC-level voltage following the motion.
- Carrier frequency ≥ 10 × highest vibration frequency to be measured.
- Advantages: frictionless, infinite resolution, linear over ± few mm to ± 25 cm, robust, good repeatability. Limits: needs AC supply and demodulator, core mass loads small structures, limited bandwidth (carrier).

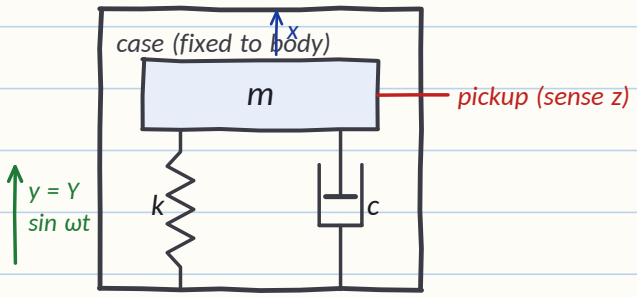
Problem 3 (LVDT)

LVDT sensitivity 100 mV/mm (after demodulation). Output amplitude 0.35 V at 20 Hz. Find the displacement amplitude and peak velocity.

$X = 0.35/0.1 = 3.5$ mm; $v = \omega X = (2\pi \times 20)(3.5 \times 10^{-3}) = 0.44$ m/s.

3. Vibration Pickups (Seismic Instruments)

Most vibration pickups use a seismic mass m on a spring k with damper c, enclosed in a case fixed to the vibrating body. The mass stays almost still in space while the case moves; the relative motion $z = x - y$ between mass (x) and case ($y = Y \sin \omega t$) is sensed (by LVDT, coil, strain gauge, piezo...).



$$m\ddot{z} + c\dot{z} + kz = m\omega^2 Y \sin \omega t$$

$$z = Z \sin(\omega t - \psi)$$

$$Z/Y = r^2 / \sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$$

$$\psi = \tan^{-1}[2\zeta r/(1 - r^2)],\ r = \omega/\omega_n$$

$$Z/(\omega^2 Y) = (1/\omega_n^2) \cdot 1/\sqrt{[\ldots]}$$

Fig 5.6 Seismic pickup and its governing equations

(Equation from Unit II, base excitation, relative motion.) The three types differ only in the range of $r = \omega/\omega_n$ where they are used.

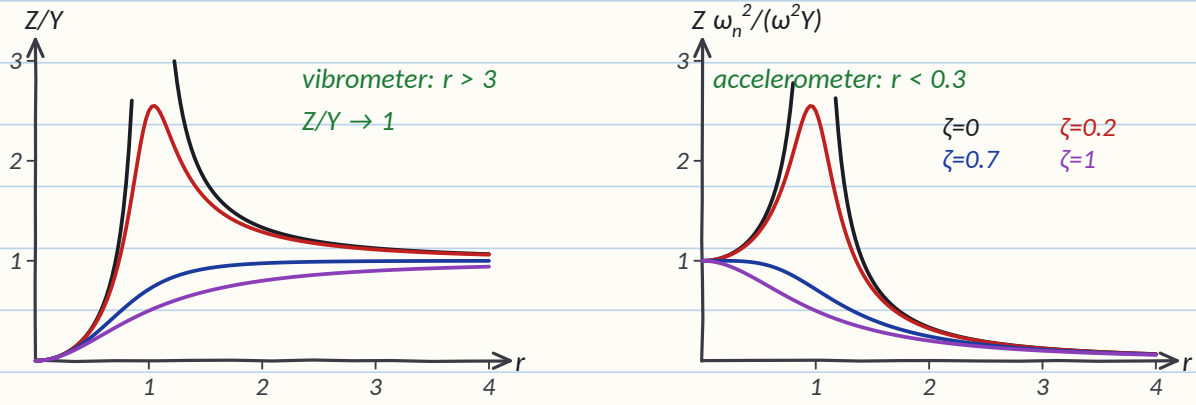


Fig 5.7 Displacement-type (left) and acceleration-type (right) response

3.1 Vibrometer (seismometer - displacement pickup)

Used for $r = \omega/\omega_n >> 1$ ($\omega >> \omega_n$). Then $Z/Y \to r^2/r^2 = 1$, $\psi \to 180°$: the mass remains stationary, relative displacement $z \approx Y$.

- Design: very low natural frequency (soft spring, large mass); useful for $\omega > \sim 3\,\omega_n$ (error small).
- Frequency range: from $\approx 3\omega_n$ upward (no upper limit). Damping $\zeta \approx 0$ - 0.7 ($\zeta = 0$ is ideal for $r > 3$, but $\zeta \approx 0.6$-0.7 gives flat response nearer ω_n).

- Drawbacks: bulky/heavy (soft spring needs large static deflection $\delta_{st} = g/\omega_n^2$; e.g. $f_n = 2$ Hz → $\delta_{st} \approx 62$ mm), cannot be used for small, light structures; large static sag; mounting orientation matters.

Problem 4 (vibrometer error)

Vibrometer $f_n = 4$ Hz, $\zeta = 0$, measures vibration at 40 Hz and reads $Z = 0.5$ mm. Find the true amplitude and error.

$r = 40/4 = 10$; $Z/Y = r^2/|1 - r^2| = 100/99 = 1.0101$ → $Y = 0.5/1.0101 = 0.495$ mm; error in reading = +1.0 %.

3.2 Accelerometer

Used for $r << 1$. Then $(1 - r^2) \approx 1$, $2\zeta r$ small: $1/\sqrt{[\ldots]} \approx 1$, so

$$Z \approx (\omega^2 Y)/\omega_n^2 = (\text{peak acceleration})/\omega_n^2 \rightarrow Z \sim \text{acceleration} \quad (4)$$

- Design: very high natural frequency (stiff spring, small mass). Frequency range from 0 (DC for strain-gauge/capacitive types) to ≈ 0.2-0.5 ω_n.
- Optimum damping $\zeta \approx 0.65$-0.7: response flat within about ±2.5 % up to $r \approx 0.5$ and phase lag nearly proportional to frequency (little phase distortion).
- Sensitivity - bandwidth trade-off: sensitivity ~ $1/\omega_n^2$. Higher ω_n (wider range) → lower output. Piezoelectric types give high ω_n and good sensitivity.
- Advantages: small, light (little mass loading), very wide range, easy to integrate once/twice to velocity/displacement. Also used for shock, impact, seismic, automotive crash tests.

Problem 5 (accelerometer)

Accelerometer $f_n = 1000$ Hz, $\zeta = 0.7$, measuring 100 Hz vibration of peak acceleration 50 m/s^2. Find error and relative displacement of the seismic mass.

$r = 0.1$; M.F. $= 1/\sqrt{[(1 - 0.01)^2 + (2\times0.7\times0.1)^2]} = 1/\sqrt{(0.9801 + 0.0196)} = 1.0002$ → error ≈ 0.02 %.

$\omega_n = 2\pi(1000) = 6283$ rad/s; $Z = (50/6283^2)(1.0002) = 1.27\times10^{-6}$ m = 1.27 μm (tiny - hence sensitive sensing needed).

3.3 Velometer (velocity pickup)

Seismic instrument with an electrodynamic sensor: a coil (or magnet) attached to the mass moves in the field of a magnet fixed to the case. Output emf ~ relative velocity $\dot{z} = \omega Z$.

- Operated at $r > 1$ (usually $r \geq 3$-4): then $Z \approx Y$, so output $e = B\,l\,\omega Y = B\,l$ (velocity of body) - a true velocity pickup; ω_n low (≈ 5-10 Hz).
- Self-generating, rugged, usable at low frequencies; output can be integrated electronically to give displacement, or differentiated for acceleration. Typical use: machinery vibration severity (velocity rms, mm/s).
- Above ≈ 1 kHz coil inductance/resonances limit the response; heavy, sensitive to magnetic fields, orientation dependent.

Pickup	Working range	Output ~	Design requirement
Vibrometer (seismometer)	$\omega >> \omega_n$ ($r > 3$)	displacement	very low ω_n, heavy, soft spring
Velometer	$\omega > \omega_n$ ($r > 1$-3)	velocity	low ω_n, electrodynamic sensor
Accelerometer	$\omega << \omega_n$ ($r < 0.3$-0.5)	acceleration	very high ω_n, light, $\zeta \approx 0.7$

3.4 Phase distortion

Real vibration is usually non-harmonic (periodic): $y(t) = Y_1 \sin \omega t + Y_2 \sin 2\omega t + Y_3 \sin 3\omega t + \ldots$ The instrument shifts each harmonic by its own phase lag $\psi_n = \psi(n\omega)$:

$$z(t) = \Sigma\, Z_n \sin(\, n\omega t - \psi_n\,) \text{ with } Z_n = |G(n\omega)|\; Y_n \quad (5)$$

- Amplitude distortion: M.F. not constant over all harmonics → harmonics amplified differently.
- Phase distortion: if ψ_n is not proportional to frequency, the harmonics are shifted by different time delays $t_n = \psi_n/(n\omega)$ → output waveform differs in shape from the input.

No distortion needs $\psi_n = n\,\omega\,t_0$ (linear phase), i.e. all components delayed by the same time t_0 (6)

- Undamped instrument ($\zeta = 0$): $\psi = 0$ for $r < 1$ and 180° for $r > 1$ - phase does not vary with frequency within each range: waveform preserved (a vibrometer with $\zeta = 0$ and all harmonics above $3\omega_n$ reproduces shape).
- With damping $\zeta \approx 0.7$: for small r, $\psi \approx 2\zeta r$ (linear!), so $t_0 = \psi/\omega = 2\zeta/\omega_n$ = constant ($\approx 1.4/\omega_n$) → minimal phase distortion over $0 < r < 1$. This is why accelerometers are built with $\zeta \approx 0.7$.
- Instruments with $\zeta \approx 0$ used close to resonance ($r \approx 1$) suffer severe distortion (phase jumps 0 → 180°). Low ζ or $\zeta > 1$ (non-linear ψ) should be avoided.

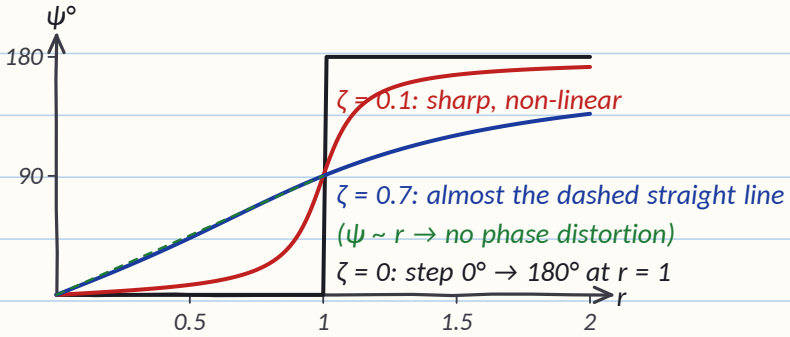


Fig 5.8 Phase lag of a seismic instrument

4. Frequency-Measuring Instruments

(a) Stroboscope

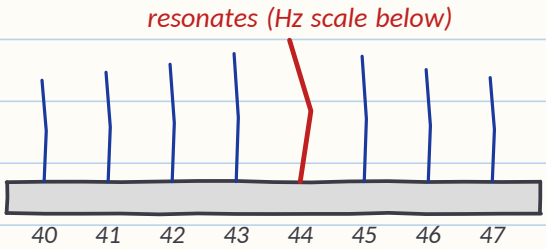
A lamp that flashes at an adjustable, calibrated rate. When flash frequency equals the vibration (or rotation) frequency, the moving part appears stationary (seen at the same position every cycle).

- Procedure: increase the flash rate from low value; the highest flash rate giving a single stationary image equals the frequency; at 2×, 3× ... that rate, one image seen; at flash = f/2, f/3 also one image (sub-harmonics) - to avoid error start from high rate and decrease.
- Non-contact, no loading of the machine, range up to ~ 20 000 flashes/min; also used to slow-motion-view vibration (flash slightly different from f).

Example: stationary single image at 1800 flashes/min; at 3600 two images (double). So speed = 1800 rpm = 30 Hz (not 3600).

(b) Vibrating reed frequency meter (Fullarton / Frahm tachometer)

A row of thin steel reeds of different lengths (each tuned to a different natural frequency, e.g. 40, 41, 42 ... Hz) fixed to a base which is pressed against the vibrating machine. The reed whose natural frequency matches the vibration resonates with maximum amplitude; frequency is read from the scale at that reed.



body vibrates at f →
reed with $f_n \approx f$ shows
largest amplitude

Fig 5.9 Vibrating reed tachometer

- Single-reed (Frahm) type: one reed whose length can be altered by a calibrated screw; tune until amplitude is maximum.
- Range ≈ 10-1000 Hz; simple, cheap; accuracy limited by reed spacing (±0.5 Hz).

(c) Oscilloscope / Lissajous figures, counters and analysers

- Lissajous figures: unknown signal on Y-plates and known generator on X-plates of an oscilloscope. Frequency ratio f_y/f_x = (number of horizontal tangents)/(number of vertical tangents) = 1:1 ellipse/circle, 1:2 figure-eight, 2:3 ...

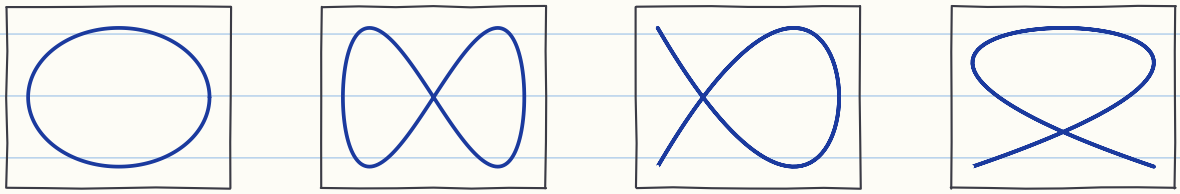


Fig 5.10 Lissajous figures for $f_x : f_y$

- Digital counters count zero-crossings (cycles per second) of conditioned signal; FFT (spectrum) analysers display amplitude vs frequency of all components (real-time, used for machine diagnostics).
- Tunable band-pass filters (octave / third-octave) and tracking filters give amplitude at selected frequency or running-speed orders.

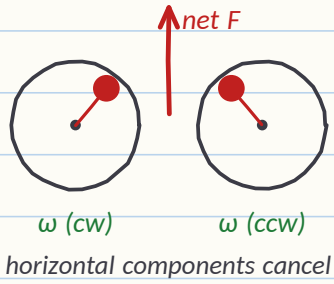
5. Vibration Exciters (Shakers)

Devices that apply controlled vibratory force/motion to a structure or specimen for: resonance testing, modal testing (FRF), fatigue and qualification (environmental) tests, calibration of pickups.

5.1 Mechanical exciters

- *Unbalanced rotating-mass (eccentric) exciter*: eccentric mass m_0, eccentricity e, speed $\omega \rightarrow$ rotating force $m_0 e\omega^2$. For a *unidirectional force* use two *counter-rotating* equal masses, horizontal components cancel:

$$F(t) = 2\, m_0\, e\, \omega^2 \sin \omega t \quad \text{(force amplitude} \sim \omega^2\text{)} \tag{7}$$



Eccentric masses on geared shafts, speed variable (motor + gears) Force changes with ω^2; useful range ≈ 5-100 Hz, force to ~ 20 kN

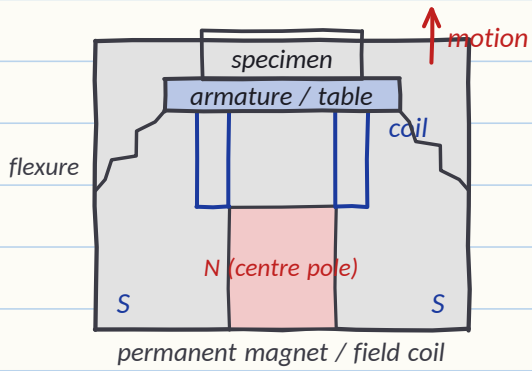
Fig 5.11 Counter-rotating mass exciter

- *Advantages:* simple, cheap, large forces at low frequency, easily mounted on structures (buildings, bridges, machine foundations). *Limits:* force depends on speed (cannot hold force constant), poor at very low frequency (force ~ ω^2), frequency range narrow, no waveform control (sinusoidal only).
- *Scotch-yoke / crank-slider / cam exciters*: a motor drives a crank, slider or cam which gives *constant displacement amplitude* (stroke) $x = X \sin \omega t$ to the specimen - used for low-frequency, large-displacement tests and fatigue; amplitude set by crank radius, frequency by motor speed (1-60 Hz).
- *Hydraulic (electro-hydraulic) exciters*: servo-valve controlled actuator; large forces and strokes at low frequency (0.1 Hz - 500 Hz) - earthquake simulators, road simulators. Also *impact hammer* and *piezo shakers* for modal testing.

5.2 Electrodynamic shaker

Most widely used for laboratory testing. Works on the *motor principle*: current i through a coil placed in a strong radial magnetic field produces force

$$F = B\, l\, i \quad (B = \text{flux density},\ l = \text{length of conductor in field}) \tag{8}$$



Signal generator → power amplifier → coil

Accelerometer on table → feedback control

Limits: stroke (low f), force (mid f),
armature resonance (high f ≈ 5-10 kHz)

Range ≈ 5 Hz - 10 kHz, force 10 N - 200 kN

Waveforms: sine, random, shock

Fig 5.12 Electrodynamic shaker (section)

- Working: amplifier drives AC current through the moving (armature) coil suspended in the radial gap of a permanent magnet (small) or DC field-coil electromagnet (large); force F = Bli moves the table and specimen at the signal frequency and waveform.
- A control accelerometer on the table gives feedback to hold the programmed acceleration; table supported by flexures, cooled by blown air.
- Advantages: wide frequency range, precise control of amplitude and frequency, any waveform (sine sweep, random, shock), low distortion.
- Limits: small stroke (≈ 25 mm) → poor at low frequency; magnetic stray field; armature resonance limits high frequency; heavy/costly; force falls if specimen heavy.

Problem 6 (electrodynamic shaker)

B = 1.2 T, length of coil in field 12 m, current 5 A; moving table + specimen mass 3 kg. Find force, acceleration, and displacement amplitude at 50 Hz.

$F = Bli = 1.2 \times 12 \times 5 = 72$ N; $a = F/m = 24$ m/s^2 (2.45 g); $X = a/\omega^2 = 24/(314.2)^2 = 0.243$ mm.

Problem 7 (counter-rotating exciter)

Two masses of 0.2 kg at e = 0.05 m, rotating at 20 Hz in opposite directions. Find force amplitude.

$\omega = 2\pi(20) = 125.7$ rad/s; $F = 2\, m_0 e\omega^2 = 2(0.2)(0.05)(15\,791) = 316$ N.

6. Applications of Vibration Measurement

Application	What is measured / how
Condition monitoring	Overall velocity rms (ISO 10816 severity: mm/s) and spectra with accelerometers on bearing housings; trending detects faults early
Fault diagnosis (spectrum)	Unbalance: 1× rpm, radial. Misalignment: 1× & 2× rpm (axial). Looseness: harmonics 1×, 2×, 3×.... Gear fault: tooth-mesh frequency = teeth × rpm with side bands. Rolling bearing: ball-pass frequencies (high freq., envelope analysis).
Modal testing	Impact hammer or shaker input + accelerometer output → FRF; peaks give ω_n, half-power gives ζ, phase gives modes
Field balancing	Vibration amplitude and phase with stroboscope/ vibration analyser; trial weights
Structural health monitoring	Strain gauges / accelerometers on bridges, buildings, aircraft - change in natural frequency indicates damage
Product testing	Shakers: sine-sweep, random, shock to qualify electronics, vehicles, packaging

One-plane field balancing (trial-weight method)

1. Measure original vibration vector O (amplitude and phase) at running speed.
2. Add trial weight W_t at a marked angular position; measure new vector O + T.
3. Effect of trial weight T = (O + T) - O (vector subtraction).
4. Correction weight $W_c = W_t \times (O/T)$ in magnitude, placed at an angle equal to the angle of trial weight, advanced by (angle of O to T) in the proper sense (so that its effect = -O).

Numerical example: O = 4 mm/s at 0°; with W_t = 10 g at 0° reading becomes 6 mm/s at 0° → T = 2 mm/s (collinear). $W_c = 10 \times 4/2 = 20$ g placed 180° from the trial position.

7. Quick Revision & Short Questions

Item	Key point
Strain gauge	GF = $(\Delta R/R)/\varepsilon \approx 2$; quarter bridge $e_0 = (E/4)GF\ \varepsilon$
Piezoelectric	$q = dF$; $e_0 = q/C$; self-generating; no DC response; wide range

Electrodynamic	$e = Blv$ (velocity); reverse: $F = Bli$ (shaker)
LVDT	Differential secondaries; $e \sim x$; phase gives direction; carrier ≥ 10 f
Vibrometer	$r >> 1$; $Z \to Y$; low ω_n; $\psi \to 180°$
Accelerometer	$r << 1$; $Z = a/\omega_n^2$; high ω_n; $\zeta \approx 0.7$
Velometer	$r > 1$; electrodynamic; output ~ velocity
Phase distortion	avoid by linear $\psi \sim \omega$; $\zeta \approx 0.7$ for small r
Frequency meters	Stroboscope, reed tachometer, Lissajous, FFT, counters
Exciters	Mechanical: $F = 2m_0e\omega^2$; electrodynamic: $F = Bli$

1. What is a transducer? A device converting mechanical vibration into a proportional electrical signal.
2. Why is a piezo accelerometer popular? High natural frequency, wide range, small, self-generating, good linearity.
3. Why can't piezo sensors measure static acceleration? The generated charge leaks away through finite insulation/input resistance.
4. Why is the LVDT output zero at the central core position? Equal and opposite emfs in the two series-opposed secondaries.
5. Difference between vibrometer and accelerometer? Vibrometer: $\omega >> \omega_n$, low ω_n, measures displacement. Accelerometer: $\omega << \omega_n$, high ω_n, measures acceleration.
6. What is phase distortion? Unequal time delays of harmonic components caused by non-linear phase-frequency response, changing the waveform shape.
7. Why is $\zeta \approx 0.7$ chosen? Widest flat amplitude response and nearly linear phase.
8. Principle of electrodynamic shaker? Force on a current-carrying coil in a magnetic field, $F = Bli$.
9. Why use counter-rotating masses? To cancel the transverse component and get a pure unidirectional force $2m_0e\omega^2$.

Important questions: (1) piezoelectric and LVDT transducers with sketches (2) theory of seismic instruments: vibrometer, accelerometer, velometer (3) phase distortion (4) frequency-measuring instruments - stroboscope, reed tachometer (5) mechanical exciter and electrodynamic shaker.